claude-windows / c6028e72-ebfd-4218-89a8-c4d7dd23be96

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\c6028e72-ebfd-4218-89a8-c4d7dd23be96\subagents\workflows\wf_f9aa1d1c-c83\agent-a62ec180ec88332d2.jsonl`
- SHA-256: `40dc958b2dcf0af067af4af64e1286cfd104ab9a3698b8df64645bc847ada6d4`
- Source modified: `2026-06-09T11:56:27+00:00`
- Imported at: `2026-07-05T16:48:23+00:00`
- project: `wf_f9aa1d1c-c83`
- session_id: `c6028e72-ebfd-4218-89a8-c4d7dd23be96`

Transcript

1. user (2026-06-09T11:53:38.475Z)

You are the deciding co-founder's technical advisor. Give COHERENT, decisive recommendations on the owner's 4 questions, integrating the research and applying the verification corrections (drop refuted/outdated claims). Respect the hard guardrails (MIT/Apache-2.0/BSD only; no paid-LLM training data; quality-first, on-device-later; multisport single conversational model). Clarify the Gemma-4-vs-ms-swift category confusion explicitly (framework vs base). Be concrete about the STARTING POINT and the next 1-2 experiments. Honest about what's unverified.

GROUNDING:
{"requirements":["GOAL (reframed): The immediate deliverable is a TRAINING FRAMEWORK + HARNESS that produces a VERY high precision/recall model for extracting HIGHLIGHTS and RALLIES from a clip. Quality is the first-order objective; everything else is secondary to model quality.","ON-DEVICE IS DEFERRED (reframe vs earlier 'on-device = immediate moat'): On-device / model-compression is a LATER goal, motivated as an upload-cost saver for power users. It is NOT the immediate target and the strategy must NOT over-index on it now. (Supersedes OWNED_MODEL_IMPLEMENTATION_PLAN.md's 'on-device = the privacy moat / immediate' framing.)","MULTISPORT FROM DAY ONE: Build for ALL net-separated racquet sports first (badminton, tennis, table tennis, pickleball, padel), with a path to eventually ALL sports. No single-sport (badminton-only) scoping.","SINGLE SPORT-AGNOSTIC MODEL: Target ONE sports-agnostic model rather than per-sport models. Collector + indexer are already sport-generic by design; the repo will be RENAMED off 'badminton'. Generic data schema stays; coach->athlete remains an optional overlay, not baked in.","SCALE COST ACCEPTED: More labels / compute / storage from multisport breadth is an explicitly accepted 'good problem' ? do not constrain the design to minimize label/compute/storage volume at the expense of breadth or quality.","CONVERSATIONAL VIDEO-QA IS THE NORTH-STAR FEATURE: upload a video -> model processes it -> user asks CONTEXTUAL questions CONTINUOUSLY (e.g. 'show me the longest rally', 'make a video of all the service faults', 'show me a clip where the rally lasted over 20 shots'). Hopefully served by a single conversational model.","PER-VIDEO STATE/BOOKKEEPING IS A FRAMEWORK REQUIREMENT: The conversational-QA loop implies durable per-video state (rally inventory, shot counts, event/fault tags, timestamps, derived clips) that the MODEL ITSELF will not hold. The FRAMEWORK must be DESIGNED FOR THIS BOOKKEEPING FROM THE START, even if the QA feature ships later.","OPEN DECISION the owner is asking for a recommendation on: is conversational-QA feasible to EMBARK ON NOW, or should it be PUNTED and revisited AFTER rally/highlight quality is solid? (The bookkeeping-from-the-start requirement holds either way.)","HARD COMMERCIAL-LICENSE GUARDRAIL ('never corrupted'): EVERY dependency ? code, training data, AND model weights ? must be usable in PRIVATE, PROPRIETARY, COMMERCIAL software. MIT / Apache-2.0 / BSD are the accepted easy fits.","REJECT viral / non-commercial / custom-restrictive licenses: explicitly reject Gemma-3 viral terms, weights trained on paid-LLM outputs, and >X-MAU caps IF truly binding for commercial use.","NEVER TRAIN ON PAID-LLM OUTPUTS: Gemini / OpenAI / Anthropic outputs must never be a training-data source (terms/copyright). Paid Gemini remains inference/serving/FALLBACK only. (Also: the existing Gemini-silver training labels in training.label_candidates must be purged before they bake in ? M0.3.)","DATA-ELIGIBILITY GUARDRAILS: Exclude minors from the training pool; per-user training data requires a SEPARATE explicit opt-in. Split data by match.","QUALITY EVIDENCE / STARTING POINT (de-risking baseline to beat): On a 6-match human-labeled corpus, the learned per-frame prototype achieves held-out

AUC 0.83-0.92 and cross-video LOVO mean 0.583 vs the heuristic's 0.480. The learned model is the chosen path BECAUSE it DECOUPLES from the multi-court 'background games' confound that the heuristic cannot escape (3 heuristic levers ? spatial gate, audio-foreground, court-region separability ? were each tested and failed this session; the problem is temporal, not spatial).","WASB PROVENANCE IS AN OPEN COMMERCIAL-CLEANLINESS BLOCKER: WASB is the shuttle/ball detector (MIT code), but the distributed checkpoint is academic-data-trained with unresolved provenance (the C3 item) ? this is a commercial-cleanliness blocker that must be resolved (document as Gen-0 prerequisite + plan own-weights, or scope an alternative).","RESOURCE/OPERATING CONSTRAINTS: Solo founder, Bangalore. Local dev box = ONE RTX 2070 Super (8GB VRAM) + WSL2 (no full-pipeline run in the dev container ? no GPU/torch/cv2/numpy there). Cloud GPU rented per-job (Modal / RunPod per the earlier study). Paid Gemini stays the fallback.","LANGUAGE/SCOPE NOTE: The reframed vision adds no new natural-language/multilingual requirement beyond English conversational-QA phrasing; 'language' as a strategic axis is limited to the conversational-QA query interface, not localization."],"tensions":["QUALITY-FIRST vs ON-DEVICE-LATER vs HARDWARE: The very-high-quality goal pushes toward larger VLM bases and cloud training, but the only local box is an 8GB RTX 2070 Super and on-device is explicitly deferred ? so the immediate model is NOT constrained to fit 8GB, yet the eventual compression goal must remain reachable. The strategy must pick a base that is both high-quality-trainable AND later-compressible without re-architecting.","MULTISPORT-FROM-START vs CURRENT EVIDENCE DEPTH: The de-risking corpus is badminton-only (WASB shuttle features, multi-court badminton confound). A single sport-agnostic model is required from day one, but the only validated signal/features and the only labeled corpus are badminton ? the feature stack (WASB trajectory) may not generalize to tennis/table-tennis/padel, and per-sport corpora don't exist yet.","SINGLE SPORT-AGNOSTIC MODEL vs COMMERCIAL-CLEAN DETECTORS: A sport-agnostic model likely needs per-sport object/ball detectors or a unified detector; WASB (badminton) already has unresolved provenance (C3), and clean MIT/Apache/BSD detectors for tennis/TT/pickleball/padel are not yet identified ? each new sport reopens the license-cleanliness guardrail.","CONVERSATIONAL-QA NORTH-STAR vs 'NEVER TRAIN ON PAID-LLM': A single conversational model that answers contextual video questions typically wants an LLM/VLM teacher, but the strongest available teachers (Gemini/OpenAI/Anthropic) are forbidden as a training source ? so the conversational capability must come from an Apache-2.0/MIT base + own data, narrowing the candidate set and raising the quality bar to clear without paid-LLM distillation.","BOOKKEEPING-FROM-THE-START vs SEQUENCING DISCIPLINE: The owner wants the framework designed for per-video QA state NOW, but the committed next slice is the async job model (single-tenant) and heavy model work is gated on scaffolding + sign-off ? designing durable QA-state bookkeeping prematurely risks building against an architecture (and a model) that doesn't exist yet (the 'embark now vs punt' question the owner is explicitly asking).","ACCEPTED SCALE COST vs SOLO-FOUNDER / SINGLE-GPU BUDGET: 'More labels/compute/storage is a good problem' collides with a solo founder on one 8GB GPU renting cloud per-job ? multisport label volume and storage are accepted in principle but still bounded by one person's labeling throughput and per-job cloud spend.","'SINGLE CONVERSATIONAL MODEL' (hoped) vs HARNESS-FIRST QUALITY GOAL: The owner hopes for one model that does detection + segmentation + conversational QA, but the harness/quality goal is concretely about a rally/highlight extractor; folding conversational QA into the same model vs keeping extraction (model) and QA (bookkeeping + retrieval/LLM layer) separate is an unresolved architectural fork.","QUALITY-FIRST ('very high precision/recall') vs HONEST EVIDENCE CEILING: Current best cross-video LOVO is ~0.583 (directional, n<=6, linear, threshold-transfer unsolved) ? a large gap from 'very high precision/recall', so the quality target may require substantially more data/features/base-model than the current de-risking artifacts demonstrate."]}

RESEARCH:

[
  {
    "dimension": "BASE MODELS for a single, multisport, conversational video-QA model that also does precise temporal localization (rally/highlight start-end)",
    "summary": "A commercially-clean base must satisfy four things at once: (1) Apache-2.0/MIT/BSD weights with NO viral/NC/MAU restriction; (2) native video understanding of long clips (full match, not a 60s window); (3) emits TIMESTAMPS for moment localization (the rally/highlight start-end requirement); (4) holds multi-turn conversation about an uploaded video (the north-star QA loop). Verified June 2026: Qwen3-VL is the only candidate that hits all four cleanly. It ships Apache-2.0 UNIFORMLY across every size (2B/4B/8B/32B/30B-A3B/235B-A22B ? no per-size license fork, unlike older Qwen generations), with a purpose-built \"Text-Timestamp Alignment\" mechanism for second-level timestamp-grounded event localization, hours-long video via 256K(?1M) context, and native multi-turn multimodal chat. Gemma 4 (released April 2026) is a genuine standard Apache-2.0 release (its Prohibited-Use Policy is non-binding guidance, not a weights restriction ? no MAU cap, no commercial bar), and adds native audio (useful for racquet-

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sport hit detection) ? but it caps video at ~60 seconds and documents NO temporal-
localization/timestamp capability, which directly disqualifies it as the PRIMARY extractor for
full-clip start-end localization. InternVL3 is license-clean (MIT project + Apache-2.0 Qwen2.5
LLM) and does multi-round video chat but has no documented native timestamp grounding. Gemma 3
is rejected (viral custom license per the existing guardrail); Gemma 3n is responsible-
commercial-license, not clean Apache-2.0, and edge-focused. Llama 4 / DeepSeek-V4 surface in
2026 roundups but Llama carries a custom community license with an MAU cap (not clean) and
neither documents racquet-grade temporal grounding. Recommendation: Qwen3-VL as the base family;
prototype the harness on the 4B/8B Instruct sizes (fit the 8GB 2070 Super for inference / small-
LoRA; rent cloud GPU for full fine-tunes and the larger sizes), keeping the option to scale to
30B-A3B/32B for the quality-first push and later distil down for the deferred on-device goal ?
all within one Apache-2.0 family so no re-architecting.",
    "options": [
      {
        "name": "Qwen3-VL (Alibaba) ? Instruct sizes 2B/4B/8B/32B + MoE 30B-A3B/235B-A22B",
        "what": "Multimodal (text+image+video) VLM series, Nov 2025, with a dedicated 'Text-
Timestamp Alignment' mechanism (beyond T-RoPE) for precise, second-level timestamp-grounded
event localization; 256K context expandable to 1M for hours-long video; native multi-turn
multimodal conversation; Thinking variants for reasoning. Active community fine-tuning + video-
grounding ecosystem (e.g. Qwen3-VL-Video-Grounding repo/Space).",
        "license": "Apache-2.0 (verified uniform across ALL sizes incl. the 235B ? no per-size
license fork)",
        "pros": [
          "Only candidate that natively emits timestamps for moment localization ? directly
matches the rally/highlight start-end requirement (Text-Timestamp Alignment, second-level
indexing)",
          "Apache-2.0 across EVERY size: clean for private proprietary commercial use; no MAU
cap, no NC clause; same license at 2B and 235B means no re-architecting when scaling quality up
or distilling down for the deferred on-device goal",
          "Handles hours-long video (256K?1M context) ? a full match clip, not a 60s window ? so
it can both localize and hold the contextual multi-turn QA the north-star feature needs",
          "Wide size ladder lets the 8GB 2070 Super run 4B/8B for inference and small LoRA
locally, while cloud-rented GPUs do full fine-tunes / larger sizes ? fits the solo-founder +
per-job cloud budget",
          "Sport-agnostic by construction (general VLM): one model can be fine-tuned across all
net-separated racquet sports, supporting the single sport-agnostic model goal without per-sport
bases",
          "Existing open fine-tuning + video-grounding tooling de-risks the training-harness
build (the immediate deliverable)"
        ],
        "cons": [
          "Base weights' training-data provenance is Alibaba's and not auditable; the Apache-2.0
license governs YOUR use, but if the owner extends the 'never train on paid-LLM' purity test to
the base's own pretraining, that cannot be fully verified (true of essentially every open VLM ?
treat as base-license-governed, not a YOUR-training-data issue)",
          "No native audio input (unlike Gemma 4) ? racquet-hit audio cues, if wanted, need a
separate audio branch or fusion",
          "Largest MoE sizes (235B) are impractical on the local box and pricey to fine-tune in
the cloud; the realistic working set is 4B?32B",
          "Quality on racquet-sport temporal localization is unproven for this corpus ? the de-
risking evidence (LOVO ~0.583) is from a separate per-frame prototype, not from this base; needs
its own fine-tune + eval"
        ],
        "fit": "strong"
      },
      {
        "name": "Gemma 4 (Google) ? E2B/E4B/12B/26B-A4B/31B",
        "what": "Google's April 2026 open-weight family; unified multimodal (text+image+video)
with NATIVE AUDIO on E2B/E4B/12B; multi-turn chat + system role; 128K?256K context. Released
specifically under standard Apache-2.0 (a deliberate break from Gemma 3's custom license).",
        "license": "Apache-2.0 (genuine standard Apache-2.0; the separate Prohibited-
Use/Intended-Use docs are non-binding policy guidance ? no MAU cap, no commercial-use bar)",
        "pros": [
          "Truly clean standard Apache-2.0 ? resolves the Gemma-3 viral-license rejection;
commercially usable in private proprietary software with no MAU cap",
```

          "Native AUDIO input (E2B/E4B/12B) ? directly useful for racquet-sport shot/hit detection, a cue the badminton work already explored",
          "Strong size ladder incl. genuinely edge-friendly E2B/E4B for the eventual deferred on-device compression goal",
          "Backed by Google with first-class tooling, quantization, and long-term support"
        ],
        "cons": [
          "DISQUALIFYING for the primary extractor: documents only ~60-second video input and NO temporal-localization/timestamp capability ? cannot emit rally/highlight start-end over a full match clip out of the box",
          "60s window also constrains the contextual multi-turn QA over a whole uploaded clip (the north-star) without heavy external chunking/bookkeeping",
          "Temporal grounding would have to be taught from scratch via fine-tuning against a base not designed to output timestamps ? higher risk/effort than Qwen3-VL's native support"
        ],
        "fit": "weak"
      },
      {
        "name": "InternVL3 (OpenGVLab) ? e.g. InternVL3-8B",
        "what": "Open multimodal series (April 2025) with strong video-understanding benchmarks and documented multi-round video conversation; built on a Qwen2.5 LLM backbone. Successor lines (InternVL3.5) add temporal-grounding gains via add-on methods.",
        "license": "MIT (project) + Apache-2.0 (Qwen2.5 LLM component) ? both clean for proprietary commercial use",
        "pros": [
          "License-clean for commercial proprietary use (MIT + Apache-2.0)",
          "Strong video-understanding benchmarks and native multi-round video chat ? fits the conversational QA loop",
          "Good size range; well-supported open ecosystem"
        ],
        "cons": [
          "No NATIVE timestamp/temporal-localization output documented in the base 3.x card ? timestamp grounding relies on add-on methods (e.g. GroundVTS) or fine-tuning, weaker out-of-box than Qwen3-VL for the start-end requirement",
          "Dual-license means tracking obligations of both the MIT project and the Apache-2.0 backbone (minor, but a compliance step)",
          "Effectively trails Qwen3-VL on the single most load-bearing requirement (native timestamp emission)"
        ],
        "fit": "medium"
      },
      {
        "name": "Gemma 3 / Gemma 3n (Google, earlier)",
        "what": "Prior Gemma generation. Gemma 3 = multimodal but under the custom Gemma license; Gemma 3n = edge/mobile multimodal (incl. audio) under a 'responsible commercial use' license.",
        "license": "Gemma custom license (Gemma 3 ? viral/custom terms) / Gemma 'responsible commercial use' (3n) ? NOT clean Apache-2.0/MIT/BSD",
        "pros": [
          "Gemma 3n is genuinely edge/on-device-oriented with audio ? relevant to the LATER compression goal",
          "Mature tooling"
        ],
        "cons": [
          "FAILS the hard commercial-license guardrail: Gemma 3's custom/viral terms are explicitly rejected by the owner; Gemma 3n's 'responsible commercial use' license is not a clean Apache-2.0/MIT/BSD",
          "Superseded by Gemma 4's clean Apache-2.0 release anyway ? no reason to adopt the restricted predecessor",
          "No advantage over Qwen3-VL on temporal localization"
        ],
        "fit": "weak"
      },
      {
        "name": "Llama 4 / DeepSeek-V4 (2026 general-LLM roundup mentions)",

```
        "what": "Large 2026 models surfaced in open-LLM roundups; Llama 4 Scout cited for very
long video context; DeepSeek-V4 variants cited as MIT with 1M context.",
        "license": "Llama = custom Llama Community License with an MAU cap (NOT clean);
DeepSeek-V4 = MIT (clean) per roundups",
        "pros": [
          "DeepSeek-V4 MIT licensing is commercially clean with very long context",
          "Llama 4 cited for long-video context windows"
        ],
        "cons": [
          "Llama Community License carries an MAU-cap / acceptable-use restriction ? fails the
'reject >X-MAU caps if binding' and clean-license guardrail; not adoptable as the base",
          "Neither documents racquet-grade NATIVE timestamp/temporal-localization for video
moment localization ? primary requirement unmet",
          "DeepSeek-V4's video-VLM + timestamp-grounding maturity is unverified vs Qwen3-VL's
purpose-built support",
          "Roundup-level sourcing only; not validated on the load-bearing temporal-grounding
requirement"
        ],
        "fit": "weak"
      }
    ],
    "recommendation": "Adopt Qwen3-VL as the base-model family. It is the only verified
June-2026 option that simultaneously satisfies all four load-bearing requirements: clean
Apache-2.0 weights (uniform across every size ? no per-size license fork, no MAU cap, no NC
clause), native long-video understanding (256K?1M context = full match clips, not a 60s window),
purpose-built timestamp emission for moment localization (its 'Text-Timestamp Alignment' giving
second-level event localization ? a direct match for rally/highlight start-end), and native
multi-turn conversational video QA (the north-star loop). Build the training harness on the
Instruct line: prototype on 4B/8B (runs on the 8GB 2070 Super for inference and small LoRA; full
fine-tunes go to rented cloud GPU), reserve 30B-A3B/32B for the quality-first push, and keep the
smaller sizes as the distillation target for the DEFERRED on-device goal ? all inside one
Apache-2.0 family so scaling up or compressing down needs no re-architecting. Keep Gemma 4
explicitly on the bench for ONE reason: its native audio input (E2B/E4B/12B) is the cleanest
path to racquet-hit audio cues if a fusion signal is later wanted ? but it cannot be the primary
extractor (60s video cap, no timestamp output). Treat InternVL3 as the license-clean fallback if
a Qwen-specific blocker emerges. Two scope notes for the harness, independent of base choice:
(1) the conversational-QA north-star is best served by keeping EXTRACTION (the fine-tuned
Qwen3-VL emitting timestamped rally/event spans) separate from the QA/RETRIEVAL layer (durable
per-video bookkeeping + query routing over those spans) ? the model emits structured spans, the
framework stores and answers over them; design that bookkeeping schema now (as the owner asks)
but PUNT shipping conversational QA until rally/highlight quality is solid, since folding QA
into the same fine-tune competes for the quality budget. (2) Base choice does NOT resolve the
WASB checkpoint provenance blocker ? that detector sits below the VLM and stays an independent
Gen-0 prerequisite.",
    "confidence": "high",
    "sources": [
      "https://github.com/QwenLM/Qwen3-VL",
      "https://huggingface.co/Qwen/Qwen3-VL-8B-Instruct",
      "https://huggingface.co/Qwen/Qwen3-VL-2B-Instruct",
      "https://huggingface.co/Qwen/Qwen3-VL-235B-A22B-Instruct",
      "https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct",
      "https://arxiv.org/abs/2511.21631",
      "https://deepwiki.com/QwenLM/Qwen3-VL/5.4-video-understanding",
      "https://github.com/PRITHIVSAKTHIUR/Qwen3-VL-Video-Grounding",
      "https://ai.google.dev/gemma/docs/core",
      "https://ai.google.dev/gemma/docs/core/model_card_4",
      "https://ai.google.dev/gemma/apache_2",
      "https://ai.google.dev/gemma/prohibited_use_policy",
      "https://ai.google.dev/gemma/docs/gemma-3n",
      "https://en.wikipedia.org/wiki/Gemma_(language_model)",
      "https://huggingface.co/OpenGVLab/InternVL3-8B",
      "https://arxiv.org/pdf/2504.10479",
      "https://llm-stats.com/models/qwen3-vl-8b-instruct"
    ],
    "caveats": [
```

      "Per-size Apache-2.0 uniformity for Qwen3-VL was verified on the 2B, 4B(referenced), 8B, and 235B HF cards plus the GitHub repo footer ? all read apache-2.0; I did not individually open every single size/edition (e.g. 32B-Thinking, 30B-A3B FP8), but the consistency across the smallest and largest plus the repo-level license makes a per-size fork very unlikely. Re-confirm the exact size you ship before release.",
      "Gemma 4's Prohibited-Use Policy was assessed as non-binding policy guidance (no MAU cap, no commercial bar) from the policy page itself; if legal certainty is required, have counsel confirm whether Google treats it as a binding condition on the Apache-2.0 grant. This does not change the recommendation (Gemma 4 is disqualified on the 60s/no-timestamp technical limits regardless).",
      "The Gemma 4 ~60-second video cap and 'no temporal localization' come from the official model card/overview as of this search; Google could extend these ? recheck if Gemma 4 becomes a serious contender. Not load-bearing for the recommendation.",
      "Base-model PRETRAINING data provenance (for ALL candidates, incl. Qwen3-VL) is not publicly auditable. The owner's 'never train on paid-LLM outputs' guardrail is satisfied at the level of YOUR fine-tuning data; whether a vendor's base pretraining touched paid-LLM outputs is unknowable and is governed by the base's license, not your training pipeline. Flag for owner's risk acceptance.",
      "Qwen3-VL's quality on THIS racquet-sport temporal-localization task is unproven ? the cited de-risking evidence (LOVO ~0.583, AUC 0.83?0.92) is from a separate per-frame prototype, not this base. The recommendation is on FIT/license/capability, not a benchmarked accuracy claim; the harness must run its own fine-tune + cross-video eval.",
      "2026 'best open LLM' roundups (Llama 4, DeepSeek-V4) are secondary/blog sources and were not individually license-verified at the model-card level; they are included only to show they were considered and ruled out on the temporal-grounding and (for Llama) MAU-cap grounds, not as adoptable candidates."
    ]
  },
  {
    "dimension": "Fine-tuning frameworks for video VLMs (SFT + RFT/GRPO for temporal grounding + conversational video-QA)",
    "summary": "FIRST, THE ORTHOGONALITY THE OWNER ASKED ME TO CLARIFY: a *framework* (ms-swift, LLaMA-Factory, axolotl, unsloth, TRL) and a *base model* (Qwen3-VL, Qwen2.5-VL, Qwen3-Omni, Gemma-4) are independent choices. The framework is the \"forge\" that runs SFT/GRPO on a GPU and emits LoRA/full weights; the base is the metal you forge. You pick ONE framework and feed it ANY Apache/MIT base, and a different base later (Qwen->Gemma) is a config swap, not a re-platforming. So this dimension does NOT pick the model; it picks the trainer. ALL FIVE leading frameworks are permissively licensed (Apache-2.0; unsloth's only AGPL piece is the optional Studio UI, trivially avoided), so the hard commercial guardrail does not eliminate any of them at the framework layer; license cleanliness is decided by the BASE WEIGHTS + DATA, not the trainer. The differentiator is capability on the four hard requirements: (1) NATIVE video training, (2) NATIVE audio training, (3) timestamp/temporal-grounding outputs, (4) RFT/GRPO with a CUSTOM reward ON VIDEO inputs, runnable on a rented A100/L40S. On those four jointly, exactly one framework clears all four in a single stack: ms-swift. It is the only framework that advertises and documents native video+audio training AND multimodal GRPO with a custom external-reward plugin executed over video rollouts (via vLLM), plus the fps/num_frames video-sampling controls you need to train timestamped grounding. The others each fail at least one hard requirement: LLaMA-Factory does native video+audio SFT but DELEGATES GRPO to a separate repo (EasyR1) that is image-only; unsloth's much-praised Vision-RL/GRPO is image-only (its value is the 8GB-box SFT, which matters LATER for compression, not now); TRL's GRPO-VLM is image-only and you would hand-build the video reward loop; axolotl explicitly labels video \"not well tested.\" NOTE ON TEMPORAL GROUNDING: emitting timestamps is primarily a BASE-MODEL + dataset-format property (Qwen2.5/3-VL align MRoPE position IDs to wall-clock time; you train on a JSON messages format with second/HMS timestamps), so the framework's job is just to expose video-frame controls and not corrupt the time encoding, which ms-swift does and is the most-documented for it. RECOMMENDATION: adopt ms-swift as the single training/RFT forge now; keep unsloth in the back pocket as the 8GB-local SFT/compression tool for the deferred on-device goal. This aligns with, and now hardens, the lean already recorded in docs/PLATFORM_ARCHITECTURE.md (which favored ms-swift but predates the RFT-custom-reward and audio requirements that make the gap decisive).",
    "options": [
      {
        "name": "ms-swift (ModelScope SWIFT)",
        "what": "Self-managed CLI/Python fine-tuning + RL + deploy framework, 600+ LLMs / 300+ MLLMs. The ONLY framework that natively trains video AND audio (Qwen3-VL, Qwen3-Omni, Gemma-4,

InternVL3.5) and runs multimodal GRPO with a custom external-reward plugin over video rollouts (vLLM-accelerated). Exposes fps/num_frames video sampling; JSON messages dataset format carries timestamps for temporal grounding. Has an eval backend (EvalScope) and args.json reproducibility, fitting the agent-runnable report-card harness goal.",
        "license": "Apache-2.0",
        "pros": [
          "ONLY framework meeting all four hard requirements in one stack: native video + native audio + timestamp/grounding controls + GRPO/RFT with a CUSTOM reward on VIDEO inputs (documented Qwen2.5-VL GRPO example uses --external_plugins + --reward_funcs + --use_vllm)",
          "Apache-2.0, clean for private/proprietary/commercial use at the framework layer; no MAU cap or NC clause",
          "Supports the exact bases on your approved list (Qwen3-VL, Qwen2.5-VL, Qwen3-Omni for audio, Gemma-4) so the framework decision does not constrain the later base choice",
          "Any model size incl. cheap-to-serve 7B/4B; runs on rented A100/L40S (your Modal/RunPod plan); multimodal GRPO examples are 6-8 GPU but SFT/LoRA fits a single 24-48GB card",
          "Custom reward plugin is the seam to reuse your existing metrics.match_intervals temporal-IoU scorer as the RFT reward (the study Gen-2 plan): single scorer spine, no metric drift",
          "Reproducibility (args.json, full_determinism) + EvalScope eval backend map onto the nanochat-style one-command report-card harness you want",
          "Active, large project (AAAI 2025; v4.0 Mar-2026), broadest model coverage of any option"
        ],
        "cons": [
          "Heaviest/most-configurable surface area, many flags; steeper first-run than unsloth notebooks",
          "Multimodal GRPO is VRAM-hungry (multiple video rollouts/sample); real video-token VRAM on 1080p/60fps clips is UNVERIFIED (same open risk flagged in the training study); re-benchmark before budgeting",
          "Audio-modality (Omni) training is less battle-tested than its video/image path; verify the Qwen3-Omni recipe before relying on audio cues",
          "ModelScope-centric defaults (some model pulls assume ModelScope hub); point it at HF mirrors for clean provenance"
        ],
        "fit": "strong"
      },
      {
        "name": "LLaMA-Factory (+ EasyR1 for RL)",
        "what": "Popular, well-documented self-managed SFT/DPO framework for 100+ LLMs/VLMs with native video (LLaVA-NeXT-Video) and audio (Qwen2-Audio, Qwen2.5-Omni) SFT. GRPO is NOT in-core; it is delegated to the sibling repo EasyR1 (a veRL fork), which is image/text-only.",
        "license": "Apache-2.0 (both LLaMA-Factory and EasyR1)",
        "pros": [
          "Apache-2.0, very approachable, best-documented onboarding; native video + audio SFT in-core",
          "Strong if your near-term need is SFT-only cold-start (Gen-2 SFT cold-start step) before any RL",
          "Large community, many tutorials, lowest ramp for a solo founder doing supervised runs"
        ],
        "cons": [
          "DECISIVE GAP: GRPO/RFT is delegated to EasyR1, which is IMAGE-ONLY (no video, no audio, no confirmed custom-reward-on-video path). So your RFT-with-temporal-IoU-reward-on-video requirement is NOT met in one stack; you would run two frameworks and EasyR1 still cannot take video",
          "Splitting SFT (LLaMA-Factory) and RL (EasyR1) doubles the harness surface and breaks the single-command report-card goal",
          "Video SFT works but the framework is not the strongest-documented for fine-grained timestamp grounding vs ms-swift Qwen-VL best-practices"
        ],
        "fit": "medium"
      },
      {
        "name": "Unsloth",

```
        "what": "Single-GPU-optimized framework (70-90% less VRAM, 1.5-2x faster) with Qwen3-VL
SFT and a celebrated Vision-RL (GRPO/GSPO) path. Trains Qwen3-VL-8B GRPO on a free Colab T4. The
RL path is IMAGE-input only.",
        "license": "Apache-2.0 core (optional Unsloth Studio UI is AGPL-3.0, avoidable; do not
ship the Studio component)",
        "pros": [
          "Best-in-class for the 8GB RTX 2070 Super: can SFT/LoRA a 7B/8B VLM locally where
others OOM, directly serving the DEFERRED on-device/compression goal",
          "Apache-2.0 core; vision GRPO/GSPO supported (image), 256K context",
          "Cheapest first runs / fastest local iteration; great for local SFT cold-start and
later quantization/compression"
        ],
        "cons": [
          "Vision-RL is IMAGE-ONLY, does not satisfy the RFT-with-custom-reward-ON-VIDEO
requirement central to the quality goal",
          "Single-GPU focus; not the multi-GPU rented-A100/L40S RFT engine for video",
          "Tightly coupled to its own patched kernels/model classes, less of a general swap-any-
base + any-reward forge than ms-swift",
          "Keep the AGPL Studio UI out of the proprietary product to stay license-clean"
        ],
        "fit": "medium"
      },
      {
        "name": "HuggingFace TRL (+ transformers)",
        "what": "The foundational RL/post-training library (GRPO, DPO, PPO, RLOO) that most
other frameworks build on. Multiple weighted custom reward functions are first-class. Its
documented GRPO-VLM path is image+text.",
        "license": "Apache-2.0",
        "pros": [
          "Apache-2.0, maximally portable, the de-facto substrate; owning the recipe here is the
ultimate anti-lock-in hedge",
          "Cleanest, most flexible CUSTOM reward API (list of reward fns, reward_weights), ideal
abstraction for your temporal-IoU reward",
          "Tight transformers integration; any HF Apache/MIT base loads directly with clean
provenance"
        ],
        "cons": [
          "GRPO-VLM is documented/tested for IMAGE inputs; video rollout plumbing (frame
sampling, video reward batching) is DIY, you become the framework author",
          "No batteries-included native video/audio dataset pipeline or fps/num_frames
ergonomics, slowest path to a working video RFT loop",
          "Lower-level: more glue code for a solo founder vs ms-swift ready video-GRPO examples"
        ],
        "fit": "medium"
      },
      {
        "name": "Axolotl",
        "what": "Production-oriented YAML-config SFT/DPO/GRPO framework with a BETA multimodal
path (Qwen2-VL, Pixtral, SmolVLM2 video, Voxtral/Gemma-4 audio).",
        "license": "Apache-2.0",
        "pros": [
          "Apache-2.0; clean reproducible YAML configs, good production ergonomics for SFT",
          "Has audio (Voxtral, Gemma-4) and some video models; LoRA/QLoRA efficient"
        ],
        "cons": [
          "DECISIVE GAP: video support is explicitly documented as not well tested at the
moment, the wrong bet for a VIDEO-FIRST very-high-quality goal",
          "Multimodal is framework-level BETA; custom-reward-on-video GRPO not demonstrated",
          "Strengths (text SFT ergonomics) are not where this project hard requirements live"
        ],
        "fit": "weak"
      }
    ],
    "recommendation": "Adopt ms-swift (Apache-2.0) as the single SFT+RFT forge now. It is the
only framework that meets all four hard requirements jointly: native video, native audio
```

(Qwen3-Omni/Gemma-4), timestamp/grounding via fps/num_frames + the JSON messages format, and GRPO/RFT with a CUSTOM external-reward plugin executed over VIDEO rollouts (vLLM); and it runs on the rented A100/L40S you already plan to use (Modal/RunPod). Its custom-reward plugin is the exact seam to wire your existing metrics.match_intervals temporal-IoU scorer in as the RFT reward (the Gen-2 plan in OWNED_MODEL_TRAINING_STUDY.md), keeping one scorer spine across heuristic-labeling, eval, and RL. The competitors each fail a hard requirement: LLaMA-Factory does native video+audio SFT but offloads GRPO to EasyR1, which is image-only (so the video-RFT goal needs a two-framework Frankenstein that still cannot take video); unsloth's Vision-RL and TRL's GRPO-VLM are both image-only; axolotl's video path is self-described as not well tested. CLARIFY FOR THE OWNER (the orthogonality you asked me to state): this decision is independent of the base-model choice; ms-swift trains Qwen3-VL, Qwen2.5-VL, Qwen3-Omni (audio), and Gemma-4 alike, so which-framework and which-base are decoupled and the base can change later without re-platforming. Because every leading framework is permissive, the commercial-license guardrail is NOT decided here; it is decided by the base WEIGHTS provenance and the training DATA (your WASB-checkpoint C3 blocker and the no-paid-LLM-labels rule live at the data/weights layer, not the trainer). Keep unsloth as the secondary, 8GB-local tool for the DEFERRED on-device/compression goal (it is the best fit for the RTX 2070 Super), and keep TRL in view as the portability hedge since ms-swift, unsloth, and EasyR1 all sit on TRL/veRL-style primitives anyway; you own the data + recipe + exported weights, so the forge stays swappable. ON CONVERSATIONAL VIDEO-QA / BOOKKEEPING: the framework choice does not force the embark-now vs punt decision; the per-video state (rally inventory, shot counts, fault tags, derived clips) is an APPLICATION/RETRIEVAL layer that NO trainer holds; ms-swift just lets you later add a QA-style SFT task over the same corpus when quality is solid. So pick the forge now, build the bookkeeping store from the start as the brief requires, and keep conversational-QA as an SFT task you can add to ms-swift later rather than a framework prerequisite.",
    "confidence": "high",
    "sources": [
      "https://github.com/modelscope/ms-swift",
      "https://swift.readthedocs.io/en/latest/BestPractices/GRPO-Multi-Modal-Training.html",
      "https://qwen.readthedocs.io/en/latest/training/ms_swift.html",
      "https://github.com/modelscope/ms-swift/issues/7326",
      "https://github.com/hiyouga/LLaMA-Factory/blob/main/README.md",
      "https://github.com/hiyouga/EasyR1",
      "https://docs.axolotl.ai/docs/multimodal.html",
      "https://www.spheron.network/blog/axolotl-vs-unsloth-vs-torchtune/",
      "https://unsloth.ai/docs/get-started/reinforcement-learning-rl-guide/vision-reinforcement-learning-vlm-rl",
      "https://unsloth.ai/docs/models/qwen3-how-to-run-and-fine-tune/qwen3-vl-how-to-run-and-fine-tune",
      "https://github.com/unslothai/unsloth",
      "https://github.com/huggingface/trl/blob/main/docs/source/grpo_trainer.md",
      "https://huggingface.co/learn/cookbook/fine_tuning_vlm_grpo_trl",
      "https://pyimagesearch.com/2025/06/16/video-understanding-and-grounding-with-qwen-2-5/",
      "https://arxiv.org/pdf/2505.20715",
      "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/PLATFORM_ARCHITECTURE.md",
      "C:/Users/avidu/Projects/badminton-highlight-indexer/docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md"
    ],
    "caveats": [
      "VRAM/token cost of multimodal GRPO on real 1080p/60fps clips is UNVERIFIED for every framework (rollouts multiply video-token cost); the documented ms-swift multimodal GRPO examples use 6-8 GPUs; re-benchmark on a rented A100/L40S before budgeting. This is the same open risk already flagged in OWNED_MODEL_TRAINING_STUDY.md, not new.",
      "Framework license is clean across the board, but that does NOT clear the COMMERCIAL guardrail; license cleanliness is decided by the BASE WEIGHTS provenance (e.g. the unresolved WASB checkpoint C3 blocker) and the TRAINING DATA (no paid-LLM labels; minors excluded). Do not let framework-is-Apache-2.0 imply the pipeline is clean.",
      "ms-swift audio (Qwen3-Omni / Gemma-4) training path is less battle-tested than its image/video path; verify the Omni recipe end-to-end before committing to audio as a training cue. Audio was NOT adopted for fusion in prior work (PR #35, audio-E1 findings), so treat audio-in-training as exploratory.",
      "Temporal-grounding/timestamp emission is mostly a BASE-MODEL + dataset-format property (Qwen MRoPE time alignment), not a framework feature; choosing ms-swift gives you the controls (fps/num_frames, timestamped messages) but the grounding QUALITY depends on the base and the data format, which must be validated separately.",

"Web specifics (version numbers, exact model lists, GPU counts in examples) are 2026 snapshots and move fast; re-confirm the live ms-swift docs/version at implementation time. Implementation remains gated on owner sign-off + PLATFORM scaffolding per the existing sequencing directive; this is a framework SELECTION, not a go-build order.",
"EasyR1 being image-only is the load-bearing reason LLaMA-Factory drops to medium; if EasyR1 (or LLaMA-Factory in-core) adds video GRPO later, re-evaluate; LLaMA-Factory SFT ergonomics are otherwise excellent for the cold-start step."
]
},
{
"dimension": "Model strategy for high-P/R rally+highlight temporal extraction (multisport, single model): fine-tuned VLM vs specialized TAL head vs both",
"summary": "The evidence says BOTH, in a deliberate sequence ? and it strongly validates the project's existing 3-generation blueprint (TAS head now ? VLM later) while sharpening the rationale. The core empirical fact: a fine-tuned VLM does NOT reach \"very high\" precision/recall at STRICT temporal boundaries. The strongest 2025-26 RL-post-trained video VLM (Time-R1 on Qwen2.5-VL-7B) reaches only R@0.7=50.1% mIoU=60.9% on Charades-STA *fine-tuned*; the same base Qwen2.5-VL-7B off-the-shelf is R@0.7=24.3%. Specialized TAL heads remain ahead on boundary precision (ActionFormer ~66.8% mAP on the harder multi-instance THUMOS14; Dec-2025 TBT-Former 68.0%). The literature consensus is explicit: VLMs \"capture coarse temporal regions rather than accurate event boundaries\" and \"struggle with absolute fine-grained temporal grounding\" ? which is exactly the rally start/end accuracy this product sells. So: the TAL/own-feature head is the right vehicle for the *quality-first* extraction goal NOW (cheap, $0 local on the 2070, already de-risked to AUC 0.84-0.91 / LOVO 0.615 at n=3), and the VLM is the right vehicle for the *conversational-QA north star* LATER ? but evidence says the QA layer should be RAG/event-index over the head's structured output, NOT a single end-to-end VLM that does both detection and conversation. A single sport-agnostic model is feasible and even beneficial: RacketVision (AAAI'26, badminton+tennis+TT) shows unified multi-sport training BOOSTS generalization +15-19% mAP, at a modest cost to per-sport pinpoint precision ? directly answering the multisport tension. Recommendation: BOTH, sequenced ? own TAL head as the quality-first extractor + teacher/label-amplifier and the honest baseline; VLM as the deferred conversational target fed by the head's event index. Embark on the bookkeeping/event-index framework NOW (it's no-regret and the QA layer needs it regardless); PUNT the end-to-end conversational VLM until extraction quality is solid and the corpus is large.",
"options": [
{
"name": "Option A ? Specialized TAL head over own feature stack (ActionFormer/ASFormer/TriDet), per-sport-agnostic, single shared head",
"what": "A tiny temporal-action-localization/segmentation head (ActionFormer 1:1 rally=instance, or ASFormer/MS-TCN2 for per-frame TAS) trained on a hand-built per-frame feature stack derived from a ball/shuttle trajectory detector (WASB-class) plus motion/audio/pose cues. One shared, sport-agnostic head; sport identity is just another input feature or a per-sport calibration, not a separate model. This is the project's de-risked Gen-0/Gen-1 path.",
"license": "MIT (ActionFormer, ASFormer, TriDet, MS-TCN2 ? use sj-li/MS-TCN2 not the Commons-Clause yabufarha repo). Trajectory detector code MIT (WASB) but the distributed checkpoint provenance is UNRESOLVED (C3 blocker) ? own-weights or a clean alternative needed.",
"pros": [
"Highest boundary precision per dollar ? specialized heads still lead VLMs at strict IoU (ActionFormer ~66.8% mAP THUMOS14 multi-instance; TBT-Former 68.0% Dec-2025). Boundary accuracy IS the product.",
"Already de-risked on THIS corpus: learned per-frame prototype AUC 0.84-0.91 held-out, LOVO mean 0.615 at n=3, overtaking the heuristic and decoupling from the multi-court confound.",
"Trains in minutes, ~4.5GB VRAM, $0 on the local 2070 Super ? fits the solo-founder/single-GPU budget; no cloud DPA needed for training.",
"Tiny (~1M params) ? trivially on-device later, satisfying the deferred compression goal without re-architecting.",
"Sport-agnostic single head is validated: RacketVision shows unified multi-sport training improves generalization +15-19% mAP. The feature stack (trajectory visibility/velocity/net-crossing) is a racket-sport invariant.",
"Doubles as the label-amplifier/teacher (confidence-gated pseudo-labels) and the honest baseline the VLM must beat ? pure upside even if the VLM later wins."
],
"cons": [
"Cannot do conversation / open-ended QA ? it only emits [start,end,label]; the north-star QA feature must live in a separate retrieval/LLM layer on top.",

          "Generalization to tennis/TT/padel is asserted-but-unproven on THIS stack: the corpus and WASB features are badminton-only today; needs per-sport trajectory detectors (clean-license ones not yet identified for all 5 sports) + per-sport labeled data.",
          "Depends on a clean trajectory detector; WASB checkpoint provenance (C3) is an open commercial-cleanliness blocker that must be resolved (own-train weights or swap, e.g. MS-TrackNetV3-class) before commercial ship.",
          "Feature engineering is per-sport-fragile at the edges (net height, ball speed regimes differ) even if the schema generalizes ? some per-sport calibration is likely."
        ],
        "fit": "strong"
      },
      {
        "name": "Option B ? Fine-tuned single end-to-end VLM (Qwen3-VL-4B/Qwen2.5-VL-7B) doing detection + segmentation + conversational QA",
        "what": "One Apache-2.0 video VLM fine-tuned to emit rally [start,end] timestamps AND answer contextual questions, via SFT cold-start then GRPO/RFT with a temporal-IoU-vs-golden reward. The owner's 'hopefully one conversational model' end-state.",
        "license": "Apache-2.0 bases exist and are clean: Qwen3-VL-4B/8B, Qwen2.5-VL-7B/32B, Gemma-4 E4B/12B-Unified (keep AMBER until raw HF LICENSE verified). BUT the standard public grounding-CoT training data (e.g. TVG-R1's 56K) is Gemini-2.5-Pro-generated ? FORBIDDEN here; you must build your own CoT labels, which is the binding cost.",
        "pros": [
          "Only path to the conversational-QA north star within a single model ? answers 'show me the longest rally', 'all the service faults', natural-language slicing.",
          "Generalizes across sports/play-types with far less per-sport feature engineering ? semantic priors transfer; one model, many sports, fewer bespoke detectors.",
          "Rapidly improving: 2025-26 RL post-training (Time-R1, MUSEG, TimeRFT) lifted Qwen2.5-VL-7B from R@0.7=24.3% (off-the-shelf) to 50.1% fine-tuned; Qwen3-VL adds explicit text-timestamp alignment for second-level indexing.",
          "Same labeled corpus feeds it via the clip?QA transcoder ? no re-labeling, consistent with the moat design."
        ],
        "cons": [
          "Does NOT hit 'very high' P/R at strict boundaries: best fine-tuned VLM is R@0.7?50% mIoU?61% on Charades-STA ? half of moments still misaligned at IoU 0.7. The literature is explicit VLMs 'capture coarse regions, not accurate boundaries.' This is a direct miss on the quality-first goal.",
          "Forbidden-teacher penalty is real and binding: the proven data recipe (~13K SFT + ~18K RL CoT) leans on paid-LLM-generated CoT labels, which this project bans ? you must originate that supervision yourself, raising the bar substantially.",
          "Cannot train locally: QLoRA of a 7B video VLM needs ~24GB+; requires rented cloud (Modal/RunPod) + signed DPA/no-train for consented footage. GRPO multiplies cost (many rollouts over long visual contexts).",
          "On-device path is heavier (4B quantized on 8GB Turing is unbenchmarked) ? defers, not deletes, the compression problem.",
          "Holding per-video QA STATE (rally inventory, shot counts, fault tags) is NOT something the model itself does reliably ? even the single-model dream still needs external bookkeeping, so it doesn't eliminate the framework requirement."
        ],
        "fit": "weak"
      },
      {
        "name": "Option C (RECOMMENDED) ? BOTH, sequenced: own TAL head as quality-first extractor + teacher/baseline; VLM deferred as the conversational target fed by a RAG/event-index over the head's output",
        "what": "Phase 1: ship the sport-agnostic TAL head as the high-P/R rally/highlight extractor (the de-risked Gen-0/1). It writes a durable per-video EVENT INDEX (rally inventory, [start,end], shot counts, ending_reason/fault tags, derived-clip refs). Phase 2 conversational QA = an LLM/agent doing RETRIEVAL over that index (AVAS-style event knowledge graph + RAG), composing answers/clips deterministically ? NOT re-deriving events. The head also serves as a confidence-gated pseudo-label teacher and the honest baseline. A fine-tuned VLM is a LATER, optional upgrade for the grounding/QA stage once the corpus is large and quality is solid ? and even then likely as a coarse-proposer refined by the head (the TimeRefine/perception-decoder two-stage pattern), not a wholesale replacement.",
        "license": "All-clean: MIT TAL heads + Apache-2.0 LLM for the QA/retrieval layer (Qwen-

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class) + own/clean trajectory weights. No paid-LLM in any TRAINING path; paid Gemini stays
inference/fallback only. The QA-layer LLM does inference-time reasoning over YOUR structured
index (allowed) ? it is not trained on paid-LLM outputs.",
        "pros": [
          "Hits the quality-first goal NOW with the boundary-accurate head, while keeping the
conversational north star reachable ? no false choice between them.",
          "RAG-over-event-index is the 2025-26 best practice for long-video contextual QA (AVAS
event knowledge graph, MemoryCard): more accurate, cheaper, and auditable than end-to-end VLM
QA, and it's deterministic for 'make a clip of all X'.",
          "Satisfies 'bookkeeping from the start': the event index IS the per-video state the QA
loop needs ? build it now as a no-regret framework artifact even if QA ships later.",
          "Decouples risk: extraction quality and conversational capability evolve
independently; you never block shipping a good extractor on an unproven VLM, and you can A/B a
VLM proposer behind the same scorer spine.",
          "Cheapest commercially-clean route: $0 local training for the head; the QA LLM is
inference-only over your data; the VLM cloud-training spend is deferred until justified by QA
volume and a $/call crossover.",
          "Same one corpus + one scorer (match_intervals / tIoU) feeds head training, TAL eval,
pseudo-labeling, AND the eventual VLM reward ? zero metric drift, the moat stays the
dataset+harness."
        ],
        "cons": [
          "Two systems to build and integrate (extractor + retrieval/QA layer) rather than one
model ? more surface area, a clear data contract (the event-index schema) is required up
front.",
          "The conversational answers are only as good as the head's event tags ? richer
semantics (e.g. 'service fault') need explicit event types the head/labels must support, adding
label taxonomy work.",
          "Defers the 'single conversational model' aesthetic the owner hopes for; if a single-
model VLM later proves it can do both at high P/R, some of the retrieval scaffolding is
superseded (but the index/corpus is reused, so low regret).",
          "Still inherits Option A's open blockers: WASB C3 provenance and multisport
feature/detector generalization must be resolved."
        ],
        "fit": "strong"
      }
    ],
    "recommendation": "Adopt Option C ? BOTH, strictly sequenced ? which is essentially a
sharpened version of the project's existing 3-generation blueprint with two course-corrections
justified by the fresh evidence. Concretely: (1) START NOW with the specialized, sport-agnostic
TAL head (ActionFormer/ASFormer-class, MIT) over the own feature stack as the QUALITY-FIRST
rally/highlight extractor ? it is the only path that credibly reaches very-high P/R at strict
boundaries on a $0 local budget, and it's already de-risked on this corpus (AUC 0.84-0.91, LOVO
0.615 at n=3). It triples as extractor, confidence-gated label-amplifier/teacher, and the honest
baseline the VLM must beat. (2) DESIGN THE PER-VIDEO EVENT-INDEX/BOOKKEEPING FRAMEWORK NOW ?
this is no-regret: the conversational-QA loop needs durable per-video state (rally inventory,
[start,end], shot counts, fault/event tags, derived-clip refs) regardless of whether QA is a VLM
or a retrieval layer. Make the index schema a first-class data contract alongside the labeled-
corpus spine (M0.1). (3) PUNT the end-to-end conversational VLM. Evidence is decisive that a
fine-tuned VLM does not hit very-high P/R at strict temporal boundaries (best-in-class
R@0.7?50%), so folding detection+QA into one model would REGRESS the core quality metric the
product sells. Build conversational QA, WHEN it ships, as an LLM/agent doing RAG over the head's
event index (AVAS-style event-knowledge-graph) ? the 2025-26 best practice for long-video
contextual QA, and deterministic for 'make a clip of all service faults.' (4) Keep a fine-tuned
VLM as a LATER, OPTIONAL upgrade for the grounding/QA stage once the corpus is large (the proven
recipe is ~13K SFT + ~18K RL examples ? and note the public grounding-CoT data is Gemini-
generated and thus forbidden here, so you must originate that supervision, which raises the
bar). If/when adopted, prefer the two-stage pattern (VLM coarse-proposes, the head/decoder
refines boundaries ? TimeRefine/perception-decoder), not wholesale replacement. (5) On
MULTISPORT: a single sport-agnostic model is feasible and beneficial ? RacketVision (AAAI'26)
shows unified badminton+tennis+TT training improves generalization +15-19% mAP at a modest
pinpoint-precision cost; keep one shared head with sport as a feature/calibration. The binding
multisport blocker is not the model ? it's clean per-sport trajectory detectors + per-sport
labeled data, and the WASB checkpoint C3 provenance, all of which must be resolved before
commercial ship. Net: do NOT start with a VLM; do NOT skip the head; build the bookkeeping now;
```

treat the VLM as a deferred QA-stage option, not the quality-first extractor.",
    "confidence": "high",
    "sources": [
      "https://arxiv.org/html/2508.10922v1 ? A Survey on Video Temporal Grounding with Multimodal LLMs (2025): VTG-MLLMs 'gradually surpassing' fine-tuned methods but 'struggle to capture fine-grained temporal dependencies' and 'precise temporal boundary prediction'; zero-shot Time-R1 R@0.5=60.8 R@0.7=35.3 on Charades-STA",
      "https://arxiv.org/pdf/2503.13377 ? Time-R1 (Qwen2.5-VL-7B, GRPO RL): FINE-TUNED Charades-STA R@0.3=82.8 R@0.5=72.2 R@0.7=50.1 mIoU=60.9 ? best-class VLM still only ~50% at strict IoU=0.7",
      "https://arxiv.org/pdf/2502.13923 ? Qwen2.5-VL Technical Report: 7B off-the-shelf Charades-STA R@0.3=67.9 R@0.5=50.3 R@0.7=24.3 mIoU=43.6 (the project's candidate base, pre-fine-tune)",
      "https://arxiv.org/pdf/2202.07925 ? ActionFormer (MIT): 66.8% avg mAP THUMOS14, 36.6 avg mAP ActivityNet 1.3 ? specialized TAL boundary-precision reference",
      "https://arxiv.org/html/2512.01298 ? TBT-Former (Dec 2025): 68.0% avg mAP THUMOS14, new SOTA, surpassing ActionFormer ? specialized heads still advancing",
      "https://arxiv.org/html/2511.17045v2 ? RacketVision (AAAI 2026, badminton+tennis+TT, 1,672 clips): unified multi-sport MS-TrackNetV3 beats single-sport +19.2% mAP tennis / +14.6% badminton, with a modest pinpoint-precision tradeoff ? validates single sport-agnostic model",
      "https://arxiv.org/html/2412.09601v1 ? TimeRefine: VLM coarse-predict-then-refine-offsets two-stage; evidence the boundary-refinement/decoder pattern is needed for fine VLM localization",
      "https://arxiv.org/html/2507.18100v1 ? TVG-R1 (EMNLP 2025 Industry): proven VLM grounding recipe = ~13K SFT cold-start + ~18K RL; public 56K CoT labels generated by Gemini-2.5-Pro (forbidden as a training source here)",
      "https://arxiv.org/html/2505.00254v3 ? AVAS: event-knowledge-graph index + RAG agentic reasoning for long-video QA ? best-practice basis for the conversational layer over an event index rather than end-to-end VLM",
      "https://arxiv.org/abs/2511.21631 ? Qwen3-VL Technical Report: explicit text-timestamp alignment for second-level video indexing (improved grounding base, Apache-2.0)"
    ],
    "caveats": [
      "Benchmark transfer caveat: Charades-STA/THUMOS14/ActivityNet are open-domain. They establish the VLM-vs-head BOUNDARY-PRECISION ceiling pattern, but absolute numbers won't map 1:1 to amateur 1080p/60fps racket footage ? rallies are more periodic/visually distinctive than generic captioned moments, so the head's relative advantage may be even larger here (favoring the recommendation), but this is inference, not measured.",
      "Multisport generalization of THIS feature stack is unproven: the de-risking corpus and WASB features are badminton-only. RacketVision validates the SINGLE-MODEL concept (and a clean MS-TrackNetV3-class detector), but tennis/TT/pickleball/padel each need clean-license trajectory detectors + labeled data not yet in hand. Pickleball/padel have far less public dataset/detector coverage than badminton/tennis/TT.",
      "WASB checkpoint C3 provenance remains an unresolved commercial-cleanliness blocker that gates BOTH options (head features and pseudo-labels) ? own-train weights or swap to a clean detector before commercial ship; a 'clean' model trained on provenance-unclear features does not launder the provenance.",
      "The forbidden-paid-LLM-teacher constraint materially raises the VLM path's cost: the entire public grounding-CoT data ecosystem (TVG-R1, etc.) is built on Gemini/GPT-generated CoT, which this project bans ? so the VLM path requires originating that supervision, a non-trivial cost that further justifies deferring it.",
      "'Very high precision/recall' should be defined operationally before the ship gate: current best cross-video LOVO ~0.615 (n=3, linear, threshold-transfer partly unsolved) is far from 'very high.' The path to high P/R is more data (n?dozens of matches across sports) + the segment-level TAL head, not a base-model swap. Set the target IoU threshold explicitly (e.g. F1@tIoU=0.5 for highlights, tighter for precise rally cuts).",
      "Qwen3-VL's grounding improvements over Qwen2.5-VL are claimed (explicit text-timestamp alignment) but not yet quantified on Charades-STA in public sources as of mid-2026 ? re-verify before any VLM-base commitment.",
      "Sequencing-discipline caveat per CLAUDE.md: this is a strategy recommendation; the committed next slice is the async job model (single-tenant). Building the event-index schema now is the no-regret action, but heavy VLM training stays gated on scaffolding + explicit owner go."
    ]
  },
  {

```
    "dimension": "Conversational Video-QA Architecture",
    "summary": "Extract once into a per-video event store; answer with a cheap query layer.
Queries are structured-numeric (text-to-SQL + clip-cut), not semantic; most of the store exists.
Build the bookkeeping now; punt the model until quality is solid.",
    "recommendation": "Adopt A: event store + session tables + a structured-query contract now;
punt the NL parser/agent until quality beats ~0.583; reject B; Gemini only a fallback.",
    "confidence": "high",
    "options": [
      {
        "name": "A event store + query layer (RECOMMENDED)",
        "what": "extract once into play_segments+events; answer with SQL + VideoCompiler clip-
cut; LLM edge-only",
        "license": "clean: SQLite/ffmpeg/MIT LLM",
        "pros": [
          "process once query many; cheap; SQL exact where VLMs hallucinate; reuses existing
tables"
        ],
        "cons": [
          "extraction quality is the ceiling; open-ended needs a VLM fallback"
        ],
        "fit": "strong"
      },
      {
        "name": "B end-to-end VLM (rejected)",
        "what": "one model answers each turn; no store",
        "license": "paid/forbidden-as-trainer",
        "pros": [
          "simplest; open-ended native"
        ],
        "cons": [
          "expensive; bad at counting; license-hostile"
        ],
        "fit": "weak"
      },
      {
        "name": "C RAG over captions",
        "what": "captions to vector/graph DB; Graph-RAG",
        "license": "paid-model risk",
        "pros": [
          "best for open-ended Qs"
        ],
        "cons": [
          "cannot count exactly; future wing only"
        ],
        "fit": "medium"
      }
    ],
    "sources": [
      "NVIDIA AVAS",
      "VideoAgent",
      "AVA 2505.00254",
      "Google ADK",
      "database.py",
      "CODE_MAP.md"
    ],
    "caveats": [
      "quality gates the feature (~0.583 LOVO)",
      "multisport semantics unvalidated",
      "single-tenant, owner-gated; fix the query contract early"
    ]
  },
  {
    "dimension": "Implementation language for the upcoming work (ML training+eval / platform /
eventual on-device), for a solo Python+FastAPI founder",
    "summary": "Recommendation: stay Python everywhere for the foreseeable roadmap, and treat
```

\"another language\" as a deferred, layer-local deployment detail ? never a rewrite. The three layers each independently point to Python or to a language choice that is structurally decoupled from your dev language, so there is no layer where a second language earns its keep against solo-founder velocity right now.\n\n(1) ML TRAINING+EVAL is effectively Python-locked and that is fine: the PyTorch/Transformers/ms-swift/ActionFormer/ASFormer stack is Python, and ms-swift is invoked as a CLI (`swift sft/eval`) ? so even the orchestrator just shells out to a subprocess. There is no non-Python alternative that retains native-video VLM training (the doc's decisive finding) or your dependency-free numpy LogisticRegression Gen-0 seed. Locking here costs nothing because training is batch/offline, not a latency hot path.\n\n(2) PLATFORM (async jobs, multi-tenant API, storage/compute backends) has no real Python \"hot path\" to escape. The heavy bytes/GPU work already runs out-of-process (ffmpeg Windows-side, WASB in WSL, cloud GPU per-job via Modal/RunPod) ? Python is the conductor, not the orchestra. The control plane is I/O-bound (DB, signed URLs, job status, polling), exactly where the GIL never bound you and where Python 3.14 free-threading + asyncio now also remove the residual CPU-bound penalty. FastAPI 0.136+ officially supports free-threaded Python. fsspec/smart_open/Arq/Hatchet (all Python, all named in PLATFORM_ARCHITECTURE.md) cover storage+queue. A Go/Rust hot-path service would buy single-digit-ms wins on requests dwarfed by 40-minute GPU jobs ? pure overhead for a solo founder.\n\n(3) ON-DEVICE is the one layer where another language eventually appears ? but it is DEPLOYMENT-only and structurally decoupled from your dev language by the export boundary. You train in Python, export to ONNX/Core ML/ExecuTorch/GGUF, and the on-device host (Swift for iOS Core ML, C++/Kotlin for ExecuTorch/Android, Rust via the `ort` crate) runs the artifact with zero Python dependency. ONNX Runtime alone exposes C++/C#/Java/Rust/JS APIs over the same model. So on-device does NOT dictate the dev language; it dictates a thin, separate, much-later runtime shell you write only if/when you ship a mobile compressed model. Per the owner's reframe, on-device is explicitly LATER (an upload-cost saver), so this is the correct layer to punt ? and punting it costs nothing architecturally because the export boundary is the seam.\n\nCross-cutting point on the conversational-QA north star and per-video bookkeeping: this REINFORCES Python-everywhere. The bookkeeping (rally inventory, shot counts, fault tags, timestamps, derived clips) is a durable relational store + retrieval/tool layer ? SQLite?Postgres + a thin query/agent layer ? which is bread-and-butter FastAPI/Python. Splitting language here would fracture the one codebase that has to hold per-video state, the eval harness, and the provider registry together.\n\nWhen a second language IS justified (write it down, don't build it now): only at on-device ship time, and only the runtime shell (Swift/Kotlin/C++/Rust host around an exported artifact) ? never the trainer, never the control plane. Premature triggers to resist: rewriting the job queue in Go \"for throughput,\" or a Rust trajectory/feature-extraction service ? both are out-of-process or numpy-vectorized already; reach for them only if profiling on real load shows a Python CPU hot path that free-threading + a C-extension (numpy/Cython/PyO3-as-a-module) can't fix.",
    "options": [
      {
        "name": "Python everywhere (all 3 layers), on-device runtime shell deferred",
        "what": "Keep FastAPI control plane, Python ML training/eval (PyTorch + ms-swift CLI + ActionFormer/ASFormer), Python platform (fsspec/smart_open storage, Arq/Hatchet jobs, Postgres + repository layer). The eventual on-device model is a Python-trained artifact exported to ONNX/Core ML/ExecuTorch; the device-side host language is chosen ONLY at mobile-ship time and is a thin, separate shell.",
        "license": "All tooling MIT/Apache/BSD-compatible (FastAPI MIT, ms-swift Apache-2.0, ActionFormer/ASFormer MIT, fsspec/smart_open BSD/MIT, Arq MIT, ONNX Runtime MIT) ? clean under the never-corrupted guardrail.",
        "pros": [
          "Maximum solo-founder velocity: one language, one toolchain, one mental model across training, eval, platform, and per-video bookkeeping",
          "ML stack has no real non-Python option that keeps native-video VLM training (ms-swift) and your existing numpy LogReg Gen-0 seed ? staying Python avoids re-implementing the de-risking artifacts",
          "Conversational-QA bookkeeping (durable per-video state + retrieval/tool layer) is exactly FastAPI/Python's strength; keeps state, eval harness, and provider registry in one codebase",
          "Python 3.14 free-threading + asyncio + FastAPI 0.136 free-threaded wheels remove the historical GIL objection for CPU-bound endpoints if it ever bites",
          "Heavy work already runs out-of-process (ffmpeg, WSL WASB, cloud GPU) ? Python is the conductor, so 'hot path in Python' is largely a non-issue",
          "On-device export boundary (ONNX/Core ML/GGUF) means the device language is a future deployment detail, not a dev-language commitment now"
        ],

"cons": [
"On-device, when it ships, still requires writing SOME non-Python runtime shell (Swift/Kotlin/C++/Rust) ? Python doesn't run well on phones; this is unavoidable regardless of dev language",
"Free-threaded Python is new (3.13 had ~40% single-thread penalty; 3.14 cut it to 5-10%) ? production maturity still settling, so treat as upside not dependency",
"Python packaging/deploy ergonomics for a self-hosted serving stack are heavier than a single Go/Rust binary (mitigated by containers + Modal-per-job)"
],
"fit": "strong"
},
{
"name": "Python core + Go/Rust hot-path service for the platform control plane / job dispatch",
"what": "Keep Python for ML + API surface, but extract job-dispatch / high-concurrency endpoints / a trajectory-feature microservice into Go or Rust 'for throughput and a single static binary.'",
"license": "Go (BSD), Rust (MIT/Apache dual) ? both clean. ONNX `ort` Rust crate is MIT/Apache. No license blocker; the objection is purely cost/benefit.",
"pros": [
"Single static binary deploy + lower memory footprint for the always-on control plane",
"Rust `ort` could serve an exported ONNX model with no Python in the serving path, if a self-hosted inference tier ever needs maximal density",
"Genuinely faster per-request latency on CPU-bound control-plane work"
],
"cons": [
"Premature for a solo founder: control plane is I/O-bound (DB, signed URLs, polling) where Python was never bottlenecked; wins are single-digit-ms against 40-min GPU jobs",
"Two languages = two toolchains, two CI paths, FFI/serialization seams, fractured per-video-QA state logic ? directly taxes the velocity the owner needs",
"No current Python hot path identified to justify it; heavy compute is already out-of-process or numpy-vectorized",
"If a Python CPU hot path ever appears, cheaper fixes exist first (free-threading, numpy/Cython/PyO3 C-extension) before standing up a second service"
],
"fit": "weak"
},
{
"name": "Rust/C++ on-device runtime shell at mobile-ship time (deferred, layer-local)",
"what": "When the LATER compression goal ships a phone-side model, write only the device-side inference host in the platform-native language: Swift+Core ML (iOS), Kotlin/C++ +ExecuTorch (Android), or Rust `ort`/C++ for a desktop power-user binary ? wrapping a Python-trained, exported artifact.",
"license": "ExecuTorch (BSD), Core ML (Apple SDK, app-distribution clean), ONNX Runtime (MIT), llama.cpp (MIT). Clean ? but verify the BASE model's on-device license (note: Gemma 3 270M GGUF is commonly cited for on-device yet is license-REJECTED per your guardrail; stay on Qwen3-VL-4B Apache-2.0 lineage).",
"pros": [
"Correct, unavoidable place for a second language: phones can't run the Python trainer; the device needs a native host",
"Fully decoupled by the export boundary ? does not touch training or control-plane code, so it never fractures the main codebase",
"Can be deferred with zero architectural cost today; the ONNX/Core ML export step is the seam, planned for but not built now"
],
"cons": [
"Real engineering surface (mobile build, quantization validation, KV-cache mgmt) ? but only when on-device is actually prioritized, which the owner explicitly defers",
"Adds a maintenance target (app + model-version compatibility) ? budget it as a distinct later phase, not now",
"Ties up scarce solo-founder time the moment it starts; gate it behind proven quality + a measured upload-cost ROI"
],
"fit": "medium"

    }
  ],
  "recommendation": "Stay Python everywhere now, and explicitly write down \"another language\" as a deferred, layer-local DEPLOYMENT decision ? not a rewrite and not a hot-path split. Concretely: (a) ML training+eval stays Python (PyTorch + ms-swift CLI as a subprocess + ActionFormer/ASFormer + your numpy LogReg Gen-0 seed); it is Python-locked and that costs nothing because training is offline/batch. (b) Platform stays Python/FastAPI; do NOT introduce Go/Rust for the control plane ? it is I/O-bound, the heavy work is already out-of-process (ffmpeg/WSL/cloud GPU), and Python 3.14 free-threading + FastAPI 0.136 cover the only remaining CPU objection. Use the Python-native pieces already chosen in PLATFORM_ARCHITECTURE.md (fsspec/smart_open, Arq?Hatchet, Postgres + repository layer). (c) On-device is the ONLY layer where a second language eventually appears, and it does NOT dictate your dev language: you train in Python and export to ONNX/Core ML/ExecuTorch/GGUF across the export boundary, then write a thin native runtime shell (Swift/Kotlin/C++/Rust-`ort`) ONLY when the LATER upload-cost-saver model actually ships. Until then, design the seam (an exportable model + a provider/ComputeBackend registry slot) and punt the shell. This keeps the conversational-QA per-video bookkeeping, the eval harness, and the provider registry in one Python codebase ? which is exactly what the quality-first, multisport, stateful-QA reframe needs. The single concrete language action now is to ensure the model architecture you train is export-clean (ONNX/Core ML-friendly ops) so on-device stays reachable without re-architecting ? a constraint on the model, not the dev language.",
  "confidence": "high",
  "sources": [
    "docs/PLATFORM_ARCHITECTURE.md (§3.2 ComputeBackend, §5 orchestration Arq/Hatchet/Temporal, §5A ms-swift training forge, fsspec/smart_open storage) ? existing Python-native posture",
    "docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md (Gen-0 ActionFormer/ASFormer MIT on local 2070, Gen-2 Qwen3-VL-4B Apache-2.0, Modal/RunPod cloud-per-job)",
    "docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md (ms-swift 'only native-video' finding; on-device VLM unbenchmarked; numpy LogReg Gen-0 seed)",
    "docs/CODE_MAP.md (heavy compute already out-of-process: ffmpeg Windows-side, WASB in WSL2; dependency-free numpy classifier.LogisticRegression; FastAPI app in backend/main.py)",
    "https://github.com/modelscope/ms-swift ? ms-swift is Python + CLI (swift sft/rlhf/eval), Apache-2.0, native video+audio training, 400+ MLLMs",
    "https://pypi.org/project/ms-swift/ ? ms-swift CLI/Python invocation surface",
    "https://onnxruntime.ai/docs/ ? ONNX Runtime cross-language APIs (C++/C#/Java/Rust via ort/JS): train in Python, run model with no Python dependency",
    "https://docs.rs/onnxruntime/ and ort crate ? Rust on-device serving of exported ONNX (YOLO-in-Rust example: Python-trained, Rust-served)",
    "https://www.edge-ai-vision.com/2026/01/on-device-llms-in-2026-what-changed-what-matters-whats-next/ ? ExecuTorch 1.0 GA (mobile), llama.cpp (desktop/proto), MLX (Apple); on-device runtime is a deployment-language choice, decoupled from training",
    "https://edgeaistack.ai/blog/edge-llm-runtime-stack-2026/ ? edge runtime stack (llama.cpp/ExecuTorch/MLX/vLLM) comparison; export-then-run model",
    "https://medium.com/@kevaldekivadiya2415/python-gil-vs-no-gil-real-fastapi-benchmarks-with-free-threaded-python-3-13-... ? FastAPI free-threaded benchmarks: CPU-bound ~8x up, I/O-bound unchanged (GIL never bound I/O)",
    "https://blog.imseankim.com/python-3-14-free-threading-jit-compiler-gil-removal-2026/ ? Python 3.14 cut free-threaded single-thread penalty to 5-10%, lock-free asyncio; FastAPI 0.136 free-threaded support"
  ],
  "caveats": [
    "The on-device second-language need is REAL but DEFERRED and layer-local: when a phone model ships you WILL write some Swift/Kotlin/C++/Rust runtime shell ? Python doesn't run on-device. The recommendation is to defer it via the export boundary, not to claim it never happens.",
    "Free-threaded Python (3.13t/3.14t) is the load-bearing reason the GIL objection is moot for CPU-bound endpoints; it is new and still maturing in production. Treat it as upside, not a hard dependency ? the platform is I/O-bound and fine on standard CPython regardless.",
    "One concrete language-adjacent constraint flows onto the MODEL (not the dev language): train with export-clean / ONNX/Core ML-friendly ops so the eventual compression target stays reachable. The tension that the immediate high-quality model isn't 8GB-constrained but must remain later-compressible is a model-architecture choice, handled in the model dimension, not here.",
    "An on-device base-model license note surfaced incidentally: Gemma 3 270M GGUF is

```
frequently cited for on-device but is REJECTED by your guardrail (viral terms). Keep on-device
bases on the Qwen3-VL-4B / Gemma-4 Apache-2.0 lineage ? a licensing item for the model
dimension, flagged so the convenient on-device default isn't accidentally adopted.",
      "If profiling on real multi-tenant load ever shows a genuine Python CPU hot path in the
control plane, escalate in this order BEFORE a second language: free-threading ->
numpy/vectorize -> a C-extension via Cython/PyO3 (still a Python module) -> only then a separate
Go/Rust service. No such hot path is identified today.",
      "ms-swift being Python-locked is not a risk: it is invoked as a CLI subprocess, so even a
non-Python orchestrator would shell out identically ? the lock-in is to the PyTorch ecosystem,
which is industry-standard and unavoidable for VLM fine-tuning."
    ]
  }
]
```


VERIFICATION:


```
[
  {
    "focus": "EXACT licenses + real 2026 existence/capabilities of every named base model (Gemma
family, Qwen2.5/3-VL per size, InternVL3) and the fine-tuning frameworks (ms-swift, LLaMA-
Factory/EasyR1, unsloth, TRL, axolotl). Verdict per load-bearing claim, flagging anything not
commercial-clean or not actually shipping.",
    "checked": [
      {
        "claim": "Qwen3-VL ships Apache-2.0 UNIFORMLY across every size
(2B/4B/8B/32B/30B-A3B/235B) with no per-size license fork.",
        "verdict": "supported",
        "evidence": "HF model cards for Qwen3-VL-2B-Instruct, -4B-Instruct, and
-235B-A22B-Instruct all read 'License: apache-2.0'; the QwenLM/Qwen3-VL GitHub repo LICENSE file
is the full Apache License 2.0. (8B/32B cards also surface as apache-2.0 in HF listings.) No
per-size fork found, unlike Qwen2.5.",
        "source": "https://huggingface.co/Qwen/Qwen3-VL-2B-Instruct ;
https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct ;
https://huggingface.co/Qwen/Qwen3-VL-235B-A22B-Instruct ;
https://github.com/QwenLM/Qwen3-VL/blob/main/LICENSE"
      },
      {
        "claim": "Older Qwen generations had per-size license forks (implying Qwen3-VL's uniform
Apache is a genuine improvement).",
        "verdict": "supported",
        "evidence": "Qwen2.5 family: all sizes Apache-2.0 EXCEPT the 3B and 72B variants, which
use the proprietary Qwen License (with 'Built with Qwen' attribution conditions). So the per-
size-fork framing is accurate for the prior generation. The finding's qualitative point holds;
just note the fork was at 3B+72B, not arbitrary.",
        "source": "https://qwenlm.github.io/blog/qwen2.5/ ;
https://huggingface.co/Qwen/Qwen2.5-72B-Instruct/blob/main/LICENSE"
      },
      {
        "claim": "Qwen3-VL has a purpose-built 'Text-Timestamp Alignment' mechanism (beyond
T-RoPE) for precise, second-level timestamp-grounded event localization.",
        "verdict": "supported",
        "evidence": "Qwen3-VL-235B HF card states it 'Moves beyond T-RoPE to precise, timestamp-
grounded event localization for stronger video temporal modeling.' Confirms native temporal-
grounding/timestamp capability as claimed. (Quantified accuracy vs Qwen2.5-VL on Charades-STA
remains unverified in public sources ? the finding's own caveat acknowledges this.)",
        "source": "https://huggingface.co/Qwen/Qwen3-VL-235B-A22B-Instruct ;
https://arxiv.org/abs/2511.21631"
      },
      {
        "claim": "Qwen3-VL was released November 2025.",
        "verdict": "outdated",
        "evidence": "Minor dating imprecision: the WEIGHTS shipped Sept-Oct 2025 (235B on Sep
23; 4B/8B on Oct 15; 2B/32B on Oct 21). November 27, 2025 is the date of the Qwen3-VL technical
REPORT (arXiv 2511.21631), not the model release. Not load-bearing for any recommendation, but
the 'Nov 2025' release label is wrong for the weights.",
        "source": "https://en.wikipedia.org/wiki/Qwen ; https://arxiv.org/abs/2511.21631"
```

      },
      {
        "claim": "Gemma 4 (released April 2026) is a genuine standard Apache-2.0 release with no MAU cap / no commercial bar; sizes E2B/E4B/12B/26B-A4B/31B; native audio on E2B/E4B/12B; caps video at ~60s with NO temporal-localization/timestamp capability.",
        "verdict": "supported",
        "evidence": "Official Gemma 4 model card confirms: license Apache 2.0; five sizes E2B(2.3B)/E4B(4.5B)/12B-Unified(11.95B)/26B-A4B-MoE(25.2B,3.8B active)/31B-Dense(30.7B); native audio on E2B/E4B/12B (ASR + speech translation, 30s max audio); video 'maximum of 60 seconds assuming the images are processed at one frame per second'; NO temporal-localization/timestamp output documented. Released April 2, 2026. All four disqualifying/qualifying technical claims verified.",
        "source": "https://ai.google.dev/gemma/docs/core/model_card_4 ; https://blog.google/innovation-and-ai/technology/developers-tools/gemma-4/"
      },
      {
        "claim": "Gemma 4's Prohibited-Use Policy is non-binding policy guidance (not a weights restriction on the Apache-2.0 grant).",
        "verdict": "uncertain",
        "evidence": "Multiple secondary/legal sources agree Gemma 4 ships under standard Apache 2.0 WITHOUT the prohibited-use-policy flow-down / unilateral-termination terms that bound prior Gemma generations (those terms 'apply to Gemma models listed in the Appendix... For Gemma 4 terms, see the Gemma 4 license'). The official model card lists 'Apache 2.0' plainly. But I did not find an authoritative Google statement explicitly characterizing the separate Prohibited-Use doc as non-binding vs the Apache grant; the finding's own caveat flags counsel confirmation. NOT load-bearing: Gemma 4 is disqualified as primary extractor on the 60s/no-timestamp technical limits regardless of how the policy binds.",
        "source": "https://ai.google.dev/gemma/docs/core/model_card_4 ; https://wcr.legal/google-gemma-license-risks/ ; https://www.mindstudio.ai/blog/gemma-4-apache-2-license-commercial-use"
      },
      {
        "claim": "Gemma 3 is under a custom/viral Gemma license (NOT Apache-2.0) and is correctly rejected by the guardrail.",
        "verdict": "supported",
        "evidence": "Gemma 3 weights are under Google's custom 'Gemma Terms of Use' with a Prohibited Use Policy, flow-down to downstream users, and a unilateral remote-restriction/termination right ? explicitly NOT Apache-2.0 (only the implementation source code is Apache). Confirms the rejection is correct.",
        "source": "https://ai.google.dev/gemma/terms ; https://techcrunch.com/2025/03/14/open-ai-model-licenses-often-carry-concerning-restrictions/"
      },
      {
        "claim": "Gemma 3n is under a 'responsible commercial use' license, NOT clean Apache-2.0/MIT/BSD.",
        "verdict": "supported",
        "evidence": "Gemma 3n is 'licensed for responsible commercial use' under the Gemma Terms of Use (not Apache-2.0): carries a Prohibited Use Policy, flow-down obligation, and unilateral termination right. Notably the Terms FOLLOW DERIVATIVES ? any model fine-tuned/distilled from Gemma stays bound. So 3n is even less clean for a proprietary fine-tune pipeline than the finding states. Rejection correct.",
        "source": "https://ai.google.dev/gemma/docs/gemma-3n ; https://ai.google.dev/gemma/terms"
      },
      {
        "claim": "InternVL3-8B is license-clean (MIT project + Apache-2.0 Qwen2.5 LLM backbone) and does multi-round video chat but has NO documented native timestamp grounding.",
        "verdict": "supported",
        "evidence": "InternVL3-8B card: 'released under the MIT License. This project uses the pre-trained Qwen2.5 as a component, which is licensed under the Apache-2.0 License.' Backbone is Qwen2.5-7B. Demonstrates multi-round video conversation (frame-by-frame) but documents NO native timestamp/temporal-grounding capability. All three sub-claims verified.",
        "source": "https://huggingface.co/OpenGVLab/InternVL3-8B"
      },
      {

        "claim": "Llama 4 carries a custom Community License with a 700M-MAU cap (NOT clean Apache/MIT) ? fails the guardrail.",
        "verdict": "supported",
        "evidence": "Llama 4 Community License: if your products exceed 700M MAU as of the release date, you must request a separate license from Meta (granted at its sole discretion), plus 'Built with Llama' attribution requirement. Custom commercial license with an MAU cap ? not clean. Correctly ruled out.",
        "source": "https://www.llama.com/llama4/license/ ; https://github.com/meta-llama/llama-models/blob/main/models/llama4/LICENSE"
      },
      {
        "claim": "DeepSeek-V4 is MIT-licensed with very long context (clean but unverified on temporal grounding).",
        "verdict": "supported",
        "evidence": "DeepSeek V4 (V4-Pro 1.6T MoE / V4-Flash 284B) released April 24, 2026 under the MIT license with open weights and a 1M-token context window. MIT-clean confirmed. Its video-VLM + timestamp-grounding maturity vs Qwen3-VL remains unverified, exactly as the finding states; included only as considered-and-ruled-out.",
        "source": "https://codersera.com/blog/deepseek-v4-release-date-features-benchmarks/ ; https://huggingface.co/deepseek-ai/DeepSeek-V4-Pro"
      },
      {
        "claim": "ms-swift is Apache-2.0 and is the only framework doing native video+audio training AND multimodal GRPO with a custom external-reward plugin over video, supporting Qwen3-VL/Qwen3-Omni/InternVL3.5/Gemma4.",
        "verdict": "supported",
        "evidence": "ms-swift GitHub: Apache-2.0; described as CPT/SFT/DPO/GRPO for 600+ LLMs and 300+ MLLMs explicitly listing Qwen3-VL, Qwen3-Omni, InternVL3.5, Gemma4. Custom reward via plugin file + '--external_plugins ... --reward_funcs'. Documented multimodal-GRPO best-practices workflow exists. The 'ONLY framework meeting all four jointly' claim is consistent with the competitor gaps verified below. (Audio-training maturity caveat stands ? Omni path less battle-tested.)",
        "source": "https://github.com/modelscope/ms-swift ; https://swift.readthedocs.io/en/latest/BestPractices/GRPO-Multi-Modal-Training.html"
      },
      {
        "claim": "LLaMA-Factory does native video+audio SFT but DELEGATES GRPO to EasyR1, which is image/text-only (no native video).",
        "verdict": "supported",
        "evidence": "EasyR1 (hiyouga, veRL-based) documents support for text, single/multiple images, and text-image combinations for Qwen2/2.5/3-VL ? with NO native video input in the core framework; video appears only via external community forks (EasyR1-video, EasyVideoR1, OneThinker). Confirms the decisive gap: the video-RFT-with-custom-reward goal is not met in one LLaMA-Factory+EasyR1 stack.",
        "source": "https://github.com/hiyouga/EasyR1 ; https://github.com/arvillion/EasyR1-video"
      },
      {
        "claim": "Unsloth's Vision-RL (GRPO/GSPO) is IMAGE-ONLY (its value is 8GB-box SFT for the deferred compression goal).",
        "verdict": "supported",
        "evidence": "Unsloth Vision-RL doc lists supported RL models (Qwen3-VL-8B, Qwen2.5-VL-7B, Gemma-3-4B, Llama-3.2-Vision inference-only) with exclusively IMAGE examples (e.g. MathVista); no video in the RL pipeline. NUANCE WORTH FLAGGING: Unsloth's SFT path DOES now support video fine-tuning for Qwen3-VL (incl. 32B/235B) per its Qwen3-VL docs ? so 'image-only' is true of the RL/GRPO path specifically, not of Unsloth SFT. The finding's framing (RL image-only; SFT/local-box value) is correct but readers should not over-read 'image-only' as covering SFT. Core (Apache-2.0) vs optional Studio UI (AGPL) split also holds.",
        "source": "https://unsloth.ai/docs/get-started/reinforcement-learning-rl-guide/vision-reinforcement-learning-vlm-rl ; https://unsloth.ai/docs/models/tutorials/qwen3-how-to-run-and-fine-tune/qwen3-vl-how-to-run-and-fine-tune"
      },
      {
        "claim": "HuggingFace TRL's GRPO-VLM path is image+text only (video rollout plumbing is DIY).",

```
        "verdict": "supported",
        "evidence": "TRL GRPO docs/cookbook: GRPO supports VLMs on text+IMAGE datasets via grpo_vlm.py; no video support documented, and even multi-image inputs are unsupported in the released version. Confirms video-RFT is DIY on TRL. Apache-2.0 confirmed.",
        "source": "https://huggingface.co/learn/cookbook/en/fine_tuning_vlm_grpo_trl ; https://github.com/huggingface/trl/blob/main/docs/source/grpo_trainer.md"
      },
      {
        "claim": "Axolotl explicitly labels video / multimodal 'not well tested' (beta).",
        "verdict": "supported",
        "evidence": "Axolotl's MultiModal/VLM docs page is titled '(BETA)' and the support 'is not well tested at the moment'; multimodal fine-tuning was introduced as a beta feature in March 2025. Video loading exists (e.g. SmolVLM2) but under the beta/'not well tested' caveat. Apache-2.0 confirmed. Correctly the weakest fit for a video-first goal.",
        "source": "https://docs.axolotl.ai/docs/multimodal.html"
      }
    ],
    "corrections": [
      "DATING (minor, not load-bearing): Qwen3-VL WEIGHTS released Sept-Oct 2025 (235B Sep 23; 4B/8B Oct 15; 2B/32B Oct 21), not 'Nov 2025'. November 27, 2025 is the technical-report/arXiv date only. The finding repeatedly labels Qwen3-VL a 'Nov 2025' release ? correct the release date if it appears in any shipped doc.",
      "UNSLOTH FRAMING (clarify, not a refutation): The finding's 'unsloth Vision-RL is IMAGE-ONLY' is correct for the RL/GRPO path, but Unsloth's SFT path now DOES support Qwen3-VL VIDEO fine-tuning (incl. 32B/235B). The recommendation (ms-swift for video-RFT; unsloth as the 8GB-local SFT/compression tool) is unaffected and is actually reinforced ? just ensure the prose doesn't imply unsloth can't touch video at all.",
      "GEMMA-3N STRONGER-THAN-STATED: Gemma 3n's non-Apache 'responsible commercial use' Terms FOLLOW DERIVATIVES (any model fine-tuned/distilled from it stays bound). This makes 3n an even cleaner reject for a proprietary fine-tune pipeline than the finding argues ? worth stating, since 3n is sometimes tempting for the on-device goal.",
      "GEMMA-4 PROHIBITED-USE-POLICY (open item, not load-bearing): I could corroborate that Gemma 4 ships under standard Apache-2.0 and that the prior-generation prohibited-use/flow-down terms are scoped to the OLD models (not Gemma 4), but I could NOT find an authoritative Google statement explicitly declaring the separate Prohibited-Use doc 'non-binding' on the Apache grant. Keep the finding's existing caveat (have counsel confirm). Does not change the recommendation ? Gemma 4 is disqualified as primary extractor on the 60s/no-timestamp limits regardless.",
      "NO COMMERCIAL-CLEANLINESS FAILURES FOUND in the recommended stack: Qwen3-VL (Apache-2.0, all sizes), InternVL3 (MIT+Apache-2.0), Gemma 4 (Apache-2.0), DeepSeek-V4 (MIT), and all five frameworks (ms-swift/LLaMA-Factory/EasyR1/unsloth-core/TRL/axolotl = Apache-2.0; unsloth Studio UI AGPL correctly flagged-avoidable) are commercial-clean as claimed. The correctly-rejected non-clean models are Gemma 3 (custom), Gemma 3n (Gemma Terms), and Llama 4 (700M-MAU cap) ? all verified non-clean exactly as the finding states. Net: the core base-model and framework recommendations (Qwen3-VL family + ms-swift forge) survive adversarial license/capability verification intact."
    ]
  },
  {
    "focus": "three",
    "checked": [
      {
        "claim": "VLM ceiling R at 0.7 near 50; off shelf 24.3",
        "verdict": "supported",
        "evidence": "Time R1 50.1; Qwen2.5 VL 24.3",
        "source": "2503.13377"
      },
      {
        "claim": "Qwen3 VL Apache plus timestamps; Gemma4 60s cap",
        "verdict": "supported",
        "evidence": "HF apache; Gemma4 60s no ts",
        "source": "HF gemma4"
      },
      {
        "claim": "RacketVision as multisport temporal model",
```

```
        "verdict": "refuted",
        "evidence": "ball tracking only not temporal",
        "source": "2511.17045"
      },
      {
        "claim": "ms swift only video GRPO",
        "verdict": "uncertain",
        "evidence": "doc image only; 7326; EasyR1 fork",
        "source": "7326"
      }
    ],
    "corrections": [
      "RacketVision is ball tracking not temporal"
    ]
  }
]
```

2. assistant (2026-06-09T11:53:41.998Z)

I'll review the key project docs referenced in the grounding to ground my recommendations in the actual repo state, then synthesize.

3. user (2026-06-09T11:53:42.927Z)

```
docs\EVALUATION.md
docs\TRACKNET_WSL_SETUP.md
docs\GOLDEN_SET_IMPLEMENTATION.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\HOW_RALLY_DETECTION_WORKS.md
docs\archives\COMMERCIAL_READINESS_REVIEW.md
docs\archives\MONETIZATION_AUDIT.md
docs\archives\README.md
docs\DEFERRED_HARDENING_PLAN.md
docs\PMF_MARKET_RESEARCH.md
docs\ROADMAP.md
docs\archives\checkpoints\2026-06-eval-baseline\README.md
docs\archives\checkpoints\2026-06-logging-reporting\README.md
docs\archives\checkpoints\2026-06-scaffolding-complete\README.md
docs\archives\decisions\DECISIONS.md
docs\archives\past_projects\low-regret-moves\LOW_REGRET_MOVES_PLAN.md
docs\archives\past_projects\README.md
docs\archives\research\AUDIO_E1_FINDINGS.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\COMMERCIALIZATION.md
docs\PLATFORM_ARCHITECTURE.md
docs\BACKLOG.md
docs\GOLDEN_SET_VENDORING.md
docs\README.md
docs\VIDEO_RECORDING_GUIDELINES.md
docs\archives\checkpoints\2026-06-strategy-foundation\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-initial\README.md
docs\OWNED_MODEL_IMPLEMENTATION_PLAN.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
docs\CODE_MAP.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\quality-iterations\README.md
```

4. user (2026-06-09T11:53:43.227Z)

```
1       # Platform Architecture ? Multi-User "Assistant Coach"
2
3       **Status:** ? Strategy / research ? for review. No code yet.
```

Date: 2026-06-07 · **Owner:** avidullu

Purpose. Map the path from today's **single-user, local** tool to a **multi-tenant "assistant coach" platform**: users upload videos, get highlights **stored and browsable by history**, and choose **where their data lives** (Google Drive / cloud bucket / local) and **where the compute runs** (our GPU / their VM). The organizing principle is to **decouple the layers** ? identity, ingestion, storage, orchestration, execution/compute, the detection+highlight pipeline, and data-serving ? behind small **pluggable interfaces**, reusing the registry pattern the codebase already uses for segmenters/providers/validators/sports/annotations.

Product framing (the north star *and* the v1 line). This is an **assistant coach**, not just a clipper ? but the strategy is **phased and unambiguous about scope**:
- **v1 (concrete, near-term):** **rally detection ? highlight-reel compilation ? per-user history ? a coach/user correction loop that feeds the golden set** ("assistant-coach *lite*"). Essentially today's pipeline made multi-user ? this is the real v1 deliverable.
- **Mid-term:** **rich rally/shot semantic metadata** (shot type, rally outcome) and the **owned video-QA model** (§5A; §6 P4?P5).
- **North-star vision (strictly future ? *not* v1):** **gait / biomechanics, per-athlete identifying templates, and deep coaching feedback under a supervising coach** (§6 P6). Inspiring, but explicitly deferred ? and the part that raises the **biometric** privacy bar (§7).

The buyer is the **academy/coach** (beachhead in [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md)); **power users self-serve**. Two consequences ripple through the design: (1) the data model keeps an **optional coach ? athlete overlay over a generic core** (§4.1) ? not baked in; (2) gait/biometrics is **special-category data**, gated and deferred (§7).

> **Competitive note (validated by research):** none of Veo / Hudl / SwingVision /
> Spiideo offer true BYO-storage or BYO-compute ? they're vendor-cloud SaaS
> (SwingVision uniquely runs *edge* compute on the user's iPhone). So **"your video,
> your storage, your compute"** is a genuine differentiator for privacy-sensitive
> academies ? and SwingVision proves user-side compute is commercially viable. All four
> share the same **teams ? athletes ? clips** data model, which we adopt (§4).

> Scope: this is a **direction-setting doc**. It does not start implementation; the
> first infra slice (an async job model) is the on-hold #16 in
> [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md). Commercial constraints come
> from [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) (C4 paid-LLM no-train; C12 SaaS
> foundations) and `archives/MONETIZATION_AUDIT.md`.

1. Current state & gaps (what we're evolving from)

Concern	Today	Gap for multi-user
Identity / tenancy	none	no users, orgs, auth, or isolation
Storage	local `filepath` string; output to `output_dir`	no abstraction; can't BYO Drive/bucket; raw video must be on the box
Data model	`videos` keyed by file **MD5**, `play_segments`, `events`, `candidate_segments` (SQLite)	no owner/athlete link; **highlights aren't a first-class persisted entity** (compile writes a file and returns a path); no per-user history
Execution	pipeline runs **in-process, synchronously** in `/api/process`	blocks; can't dispatch to a user's VM/GPU; no job status
Pipeline	**already pluggable** (segmenter/provider/validator/sport/annotation registries)	reuse as-is ? this layer is the model for the rest
API	`/api/process`, `/api/query`, `/api/compile`, `/api/config`; no auth	needs auth, per-tenant scoping, async job + history endpoints

Grounding: `backend/infrastructure/database.py` (schema + `PRAGMA user_version` migration ladder, v1), `backend/main.py` (endpoints, sync flow), the registry pattern in `backend/pipeline/segmenters/base.py::SegmenterRegistry` and `backend/providers/base.py`.

2. Target layered architecture

Decouple into layers with a clean **control-plane vs data-plane** split ? the orchestrator coordinates *metadata and jobs*; the heavy *bytes and GPU work* can live elsewhere (even in the user's own cloud).

```
                ????????????????????????????????????????????????
   CONTROL      ?  Presentation / API  (auth, per-tenant)       ?
   PLANE        ?  Identity & Tenancy  (org?coach?athlete?user)  ?
                ?  Orchestration / Jobs (submit?status?result)   ?
                ??????????????????????????????????????????????????
                                ? dispatch       ? metadata
                ?????????????????????????  ???????????????????????
   DATA         ?  Execution / Compute  ?  ?  Data / Serving     ?
   PLANE        ?  (local | BYO-VM |    ?  ?  (per-tenant DB:    ?
                ?   hosted GPU)         ?  ?   videos, segments, ?
                ?   runs the Pipeline:  ?  ?   highlights, jobs, ?
                ?   detect ? highlight  ?  ?   history)          ?
                ?????????????????????????  ???????????????????????
                            ? read input / write artifacts
                ?????????????????????????????????????????????????
                ?  Storage  (local upload | Google Drive | S3 |  ?
                ?  GCS | Azure)  ? raw video + highlight outputs ?
                ??????????????????????????????????????????????????
```

The two **new pluggable seams** are **Storage** and **Compute/Execution**; everything else either exists (pipeline) or is net-new infra (identity, orchestration, serving). A **model/training layer** ? the owned generative video-QA model (§5A) ? sits atop these, fed by Storage (training corpus), Data (labels), and Compute (training + inference), and is delivered through the existing **AI-provider registry with a paid fallback**.

3. Pluggable abstractions (extend the existing registry pattern)

The codebase already proves the model: an `ABC` + a `Registry` singleton with `register/get/list_available` (e.g. `SegmenterRegistry`). Add two more registries with the same shape so they're plug-and-play and testable in isolation.

3.1 `StorageBackend` registry
Replace the bare `video_path: str` that flows through the pipeline with a **`StorageRef`** (a backend id + a key/URI), resolved by a backend.

```python
class StorageBackend(ABC):
    name: str
    def open_read(self, ref: StorageRef) -> BinaryIO: ...      # stream large video
    def download_to_local(self, ref: StorageRef, dst: str): ...  # for ffmpeg/cv2/WSL
    def upload(self, local_path: str, ref: StorageRef): ...     # persist highlight/artifacts
    def signed_url(self, ref: StorageRef, ttl: int) -> str: ... # hand to LLM / browser
    def stat(self, ref: StorageRef) -> StorageStat: ...
    def list(self, prefix: StorageRef) -> list[StorageRef]: ...
```

- **Recommended implementation:** build the backend over **`fsspec`** (one matrix ? local / S3 (`s3fs`) / GCS (`gcsfs`) / Azure (`adlfs`) + Google Drive via `gdrivefs`, Dropbox via `dropboxdrivefs`), reaching for **`smart_open`** as the robust large-MP4 byte-streaming primitive. Storage becomes one more registry. (Treat Drive/Dropbox as add-ons ? less battle-tested than `s3fs`/`gcsfs`; cover with integration tests.)
- **Backends:** `local` (upload dir), `gdrive`, `s3`, `gcs`, `azure`. **BYO-storage**: the tenant registers their own bucket/Drive; raw video **never leaves their account**.

- **Auth ? never persist users' long-lived keys.** For buckets, have the user create an IAM role you **`AssumeRole`** into, scoped to one bucket/prefix, and mint **presigned URLs** so raw video moves **browser ? the user's bucket directly, never through our servers**. For Drive, OAuth with the narrow **`drive.file`** scope; for academies, a **service account with domain-wide delegation**. Store any per-tenant creds encrypted.
- **Why it matters for the pipeline:** ffmpeg/OpenCV/WSL need a **local file**, while the LLM provider can take a **signed URL**. The backend exposes both `download_to_local` (for compute) and `signed_url` (for the provider/browser), so the pipeline stops caring *where* the bytes live.
- **Migration path:** introduce `StorageRef` alongside the current local path; a `local` backend wraps today's behavior so nothing breaks; adapters land incrementally.

3.2 `ComputeBackend` / `ExecutionBackend` registry
Where the pipeline actually runs for a given job.

- **`local`** ? in-process / local subprocess (today's behavior; dev + self-host).
- **`hosted`** ? our GPU VM pool (the SaaS default).
- **`byo_worker`** ? the tenant runs a **pull-based worker** (a small agent) on their own VM/GPU. **Critical security property:** the worker *pulls* jobs scoped to its tenant from the control plane, pulls input from the tenant's **own** storage, runs the pipeline, and writes artifacts + results back ? so **the orchestrator never holds the user's cloud credentials** and needs **no inbound access** to their network. This mirrors the **self-hosted CI runner** pattern (GitHub Actions runners use a 50s outbound-only HTTPS long-poll). This pull-based runner is the **cheapest, most credential-safe v1** for BYO-compute.
- For users who want the platform to **provision GPUs in their *own* cloud account on demand** (an academy with an AWS account but no standing GPU box), add **SkyPilot** (BYOC: everything launches in the user's accounts/VPCs; creds stay with them). **Modal** is the managed fallback for users wanting zero infra (accepting that raw video then flows through Modal).
- **?? Operational / support burden (a real adoption risk).** For the primary beachhead ? Indian badminton **academies/coaches** ? asking them to *run and maintain a worker on their own VM/GPU* is a meaningful ongoing load (updates, monitoring, firewall/NAT, the "it stopped after our IT change" ticket). So: **default academy users to `hosted`**, and treat `byo_worker`/SkyPilot as a **power-user / enterprise / privacy-sensitive tier**, not the academy default. If/when BYO ships, plan for **auto-updating, self-healing agents + health heartbeats + clear worker status in the UI**, and budget the support cost.
- **Middle tier to consider ? "zero-infra BYO":** a managed-but-isolated path where the **storage** stays in the user's bucket (signed URLs) while compute runs on **our** GPUs in a **strictly isolated, per-tenant, ephemeral sandbox** (nothing persists our side). This gives the BYO *privacy* benefit (raw video never persists on our infra) **without** the user running a worker ? a pragmatic default between full-hosted and full-BYO.

The pipeline code is unchanged; only *where it's invoked* and *how it gets bytes* vary.

3.3 Reuse, don't reinvent
The existing `segmenter` / `provider` / `validator` / `sport` / `annotation` registries are the proven precedent and stay as the **pipeline** layer. The `annotation` registry + `candidate_segments` table are also the natural seam for the **coach-correction / golden -set** loop in the assistant-coach phase (§6, P4).

4. Multi-tenant data model & serving layer

4.1 Entities ? generic core, with the coach model as an optional overlay
The foundational schema is deliberately **domain-neutral** so we don't overfit to the coach mindset ? the coach/athlete framing is a **market-fit hypothesis that will iterate**

(see C13 / PMF), not a load-bearing assumption. **Generic core:**
```
User ??< VideoCollection ??< Video ??< Highlight ??< HighlightMetadata
                                 ???< PlaySegment ??< Event
Video / Highlight ??< Job (processing record: status, compute backend, timings)
```
- **Generic & re-targetable:** a `VideoCollection` groups a user's videos (a season, an event, a "folder") with no assumption of a coach or academy ? a power user, a coach, or a parent are all just `User`s with collections.
- **Highlights are first-class** (today compile only writes a file): a `highlights` table (segment ids + output `StorageRef` + params + created_at) powers "see past videos and highlights by history," and **`HighlightMetadata`** carries the **rich semantic tags** (shot type, rally outcome, etc. ? see §6 metadata stage) plus titles / ratings / notes.
- **`video_id` stays the content MD5** (dedup within a tenant), scoped by `tenant_id`.
- **Coach overlay (optional, additive):** for the academy/coach GTM, layer `Org/Academy ? Coach ? Athlete` *on top* (e.g. a `Membership`/role table + an `athlete_id` tag on collections/videos) **without** baking it into the core tables ? so the schema stays usable for solo power users and the coach model can evolve, or be dropped, as PMF dictates.
- **History/serving API:** list collections/videos/highlights per user (and, with the overlay, by athlete/coach/date); **query highlights by `HighlightMetadata`** (e.g. shot type / outcome); re-stitch from stored segments without reprocessing.

4.2 Tenancy isolation ? options & recommendation

Model	Isolation	Ops cost	Best for
Shared DB + `tenant_id` (row-level, ideally Postgres **RLS**)	logical	low	**Hosted SaaS, many small tenants** ? *recommended default*
Schema-per-tenant	medium	medium	tens?hundreds of mid tenants
DB-per-tenant (incl. **SQLite-per-tenant**)	strong/physical	higher at scale	**self-hosted single-academy**, or high-isolation enterprise

Recommendation: for the hosted product, **migrate SQLite ? Postgres with shared-DB + `tenant_id` + Row-Level Security**; keep **SQLite-per-tenant** as the **self-hosted / single-academy** deployment (it reuses today's code almost verbatim). Either way, **abstract DB access behind a repository layer now** so the engine is swappable, and graduate the `PRAGMA user_version` migration discipline to **Alembic** under Postgres.

5. Orchestration / async jobs

This is the serving layer that makes multi-user real. It is the **multi-user superset of [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md) #16** (async job model) ? build that first, single-tenant, then generalize.

- **Job record** (`jobs` table): `id, tenant_id, video_id, type (index|compile|analyze), status (queued|running|done|failed), progress, compute_backend, result_ref, error, timestamps`. `/api/process` returns **202 + job_id**; `/api/jobs/{id}` polls.
- **Idempotency** keyed on `(tenant_id, video_id, pipeline_version)` ? reuse the content-hash discipline already in `compute_video_id`.
- **Dispatch:** the orchestrator enqueues; a worker (hosted or `byo_worker`) claims jobs for its tenant, resolves input via the tenant's `StorageBackend`, runs the pipeline, writes artifacts + DB rows. **Observability reports** (already per-video) become the job artifact surfaced in history.
- **Queue/worker backend ? a decision to prototype in #16, *not* a default here.** The *requirements* are firm ? **idempotency**, **resumability** (a 40-min GPU job mustn't restart on a worker blip), **progress states** ? which rule out fire-and-forget Celery/RQ for the long jobs. The *choice* should be made by prototyping in the

  [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md) #16 slice, weighing:
  - **Start simplest:** **Arq** (async-native, natural FastAPI fit) or even a **plain
    Postgres-backed queue + worker** ? likely sufficient for the *first* single-tenant
    async slice, with the lowest operational complexity and the easiest debugging.
  - **Durable-execution upgrade if/when needed:** **Hatchet** (Postgres-backed, per-step
    retries/timeouts/concurrency; pairs neatly with the pull-based worker) ? attractive
    because it adds **no new datastore** once we're on Postgres, but it *is* a new
    dependency + operational/debugging surface; adopt only when long-job resumability/
    observability demands it.
  - **Heaviest:** **Temporal** ? most mature durable workflows, operationally heavy;
    reserve for genuinely complex multi-stage pipelines at scale.

  Record the decision criteria in #16: operational complexity, maturity, **debugging
  experience for long GPU jobs**, and whether durability is worth a new dependency.
  **Bias: start with the simplest thing that gives retries + status; graduate only on
  evidence.**

5A. Owned generative model (the video-QA assistant) & the data flywheel

The north-star capability: an **owned, self-hostable generative video model** ?
fine-tuned on sports video (including **consented, user-contributed** footage) ? that
answers questions about rallies, shots, and technique. This is the conversational
brain of the "assistant coach." Crucially, it does **not** replace the known stack; it
is introduced **behind the same abstraction, with the paid API as an always-ready
fallback**.

> **The moat is *not* the model.** Anyone can download Qwen-VL/Gemma. The defensible
> assets are **(a) the consented, badminton-specific video?QA dataset** and **(b) the
> eval harness**. Build and message *those* ? not "we own an LLM."

> **? Committed guardrail (legal, non-negotiable).** We respect the licensing findings in
> full. **Paid frontier models (Gemini / OpenAI / Anthropic) are used for *inference /
> serving / fallback* only ? never as a *training* data source.** Harvesting their outputs
> to train our owned model would breach their "no competing model" terms (a terms/copyright
> violation), so it is prohibited in both engineering and legal process. **Training teachers
> are open, Apache-2.0 models only** (Qwen-VL / InternVL / Gemma 4). Detail in the
> distillation constraint below.

Core principle: AI is a pluggable provider with a paid fallback
The codebase already abstracts the cloud AI behind `provider_registry` /
`AIVideoProvider` (`backend/providers/`). Make that a **first-class platform principle**:
- Every generative capability routes through the **provider registry**.
- The **owned model** plugs in as a new provider (`owned_vlm`); **Gemini (paid, via
  Vertex AI)** stays registered as the **fallback** (failover on error/low-confidence,
  or as the quality ceiling/teacher reference).
- This lets us **ship today on the known paid stack** and **swap in the owned model when
  its quality clears a measured bar** ? with zero architectural churn and the paid path
  always one config flip away. (Mirrors the COMMERCIALIZATION C4 A/B/C strategy: the
  owned model is the long-run "(A) eliminate the per-call dependency"; paid is "(C)".)

The model / training layer (new sub-layer, fed by the scaffolding)
```
Data curation        Training              Eval                 Serving
(consented user vid  (fine-tune an open,   (extend the existing (self-host inference
 + golden set +      commercially-usable   eval harness to QA   as a provider /
 rally/segment/event base VLM via          quality + the        ComputeBackend;
 annotations ?       LoRA/QLoRA)           regression gate)     paid fallback)
 video-QA pairs)
```
This layer **reuses every other layer**: **Storage** holds the training corpus;

Data supplies labels (the `annotation` registry + `candidate_segments` + golden set are already the labeling seam); **Compute/Orchestration** runs training and inference jobs (even **BYO-compute** can serve inference on the tenant's GPU); the **eval harness** already in `backend/eval/` extends to grade QA quality. **In other words, the rock-solid scaffolding is exactly what makes the model flywheel possible** ? which is why scaffolding comes first (see §6).

Base model & training (Apache-2.0 only ? licensing is decisive)
Recommended near-term base: `Qwen2.5-VL-7B-Instruct` ? Apache-2.0, native long-video (dynamic-FPS sampling, temporal mRoPE), mature LoRA tooling. Upgrade path **Qwen3-VL-4B/8B (Apache-2.0)**; **Gemma 4 multimodal (Apache-2.0, 2026)** is a clean Google-stack alternative to evaluate.
- ?? **Lock to Apache-2.0-clean bases** so you can fine-tune, **keep weights private, gate behind subscriptions, and never owe open-sourcing or branding**. (Note Qwen2.5-VL is Apache **only at 7B/32B** ? 3B/72B use the custom Qwen license.)
- ? **Reject for the flagship:** **Gemma 3** ? custom terms with a **viral derivative clause** (anything fine-tuned on it, *or even trained on its synthetic outputs*, stays permanently bound to Google's *updatable* Prohibited-Use Policy); **Gemma 4's switch to Apache-2.0 removes this**. **Llama 3.2 Vision** ? custom license forces a "Llama-?" model name + "Built with Llama" badge and a 700M-MAU ceiling. **`gpt-oss`** ? Apache-2.0 but **text-only**, so usable only as an optional text-reasoning layer, never the video encoder.
- **Training method:** **QLoRA** (? r=64, ?=128) on the 7B ? fits one prosumer GPU and avoids catastrophic forgetting of general video skills. Build the dataset by converting the existing rally/shot/segment annotations + golden set into templated **clip?QA pairs**, expand synthetically with **open-model teachers only** (see constraint below), and hold out a **golden QA eval** that gates every fine-tune. A few hundred gold pairs ? **5k?20k** curated is a realistic band for this narrow domain.
- **Serving economics:** a 7B VLM ? 14 GB VRAM (FP16) / ~3.5 GB (INT4); a ?16 GB GPU is a comfortable serve target (vLLM for throughput). Self-hosting wins **only** once you have sustained video-QA volume **+** a defensible consented dataset **+** real per-call API cost pain ? below that, the paid API is cheaper than a standing GPU and the second stack is overhead. Hence: **paid-primary now ? owned-primary / paid-fallback later.**

Base-model licensing + training-data policy are tracked as **COMMERCIALIZATION C14**.

Training backend ? managed service vs self-managed (the fine-tune "forge")
Where you run the LoRA/RL fine-tune. Weigh on **video support, weight export/ownership, smallest servable model, RL support, no-train DPA, and infra burden**.

Decisive finding (researched 2026-06): managed fine-tuning services are **image-based** (JSONL + base64 images) ? for video you pre-sample frames and feed them as image sequences. **Only the self-managed `ms-swift` explicitly advertises *native* video (+audio) training.** So for true video-QA, a self-managed framework is the strongest technical fit ? and it also lets you fine-tune the **small 7B** (cheap to serve) and keep full ownership. Managed services trade some of that for zero infra.

Option	Type	Video / VLM support	Smallest VL	RL	Export weights	Best for
ms-swift (ModelScope)	Self-managed (rented/own GPU)	**Native video+audio**; Qwen3-VL/Omni, 300+ MLLMs	any incl **7B**	? DPO/GRPO	? full ownership	**? video-QA; slots into BYO-compute (§3.2)**
LLaMA-Factory	Self-managed	Qwen3-VL + 100+ VLMs; video datasets	any incl 7B	?	?	popular, well-documented
Unsloth	Self-managed (1-GPU optimized)	Qwen3-VL LoRA (vision+lang; frames)	any incl 7B	? GRPO	?	cheapest first runs

Tinker (Thinking Machines)	Managed API (low-level)	Qwen3-VL **30B-A3B+ only**; image input (video **unverified**)	30B-A3B MoE	? PPO/GRPO, custom loss	? adapters + ckpt; no-train ToS	zero-infra RL/research at MoE scale
Together AI	Managed (tune + serve)	Qwen3-VL / Gemma-3 / Llama-4; image data	verify	partial	? adapters downloadable	managed Qwen3-VL + serving
Fireworks AI	Managed (tune + serve)	Full Qwen2.5-VL (3B?72B SFT; LoRA@32B + LoRA-upload)	**7B**	partial	?? verify export (lock-in risk)	managed *small*-VL + serving
Predibase (Rubrik)	Managed **+ VPC self-host**	VLM instruct-tune; **LoRAX** multi-adapter	verify	? (VPC)	? **RFT/GRPO** flagship	RL "creativity" + multi-adapter + privacy
~~OpenAI GPT-4o / Gemini tuning~~	Closed	?	?	?	? no export	**non-fit ? violates "own the model"**

(Enterprise clouds ? Databricks Mosaic / SageMaker / Vertex on open models ? give full control but are only worth it if you're already on that stack.)

Recommendation:
- **Early iteration / true video-QA ?** self-managed **`ms-swift`** (or LLaMA-Factory / Unsloth) on a **rented GPU** (RunPod / Lambda / Modal / SkyPilot-to-your-cloud): native video, any size incl. **7B** (cheap to serve), full ownership, and it drops into the **ComputeBackend / BYO-compute** seam (§3.2). **Likely a better fit than Tinker** for the video requirement *and* serving economics.
- **Managed, no GPU ops ?** **Together AI** (Qwen3-VL + downloadable adapters + serving) or **Fireworks** (full Qwen2.5-VL incl. 7B + serving) ? confirm frame/video support and adapter export.
- **If RL "coaching creativity" + per-sport multi-adapter is central ?** **Predibase** (RFT + LoRAX, VPC self-host for the biometric/privacy posture) or **Tinker** (lowest-level RL primitives).
- **Tinker** remains a strong zero-infra RL-research option with clean weight export ? just not the cheapest path to a *small video* model (its VL floor is 30B-A3B).
- **Portability is the hedge:** you own the **data + recipe + exported weights**, so the backend is swappable ? same ethos as the provider/registry pattern. Don't get locked in.

Verify before committing (any backend): (1) **native video / frame-sequence** ingestion + max frames per example; (2) **weight export** + a **commercial DPA** covering consented, minors-excluded video (biometric, §7); (3) the **inference plan** for the size you train (a self-managed **7B** is far cheaper to serve than a 30B-A3B MoE).

Training harness ? make it agent-runnable (nanochat-style operability)
The goal: **an agent (or CI) can run a fine-tune on a fresh batch of golden-labeled video and get a trustworthy ship / no-ship verdict, unattended.** Karpathy's **nanochat** is the reference for that *operability* ? one MIT repo, one `speedrun.sh` that runs tokenizer?train?eval?serve, a single complexity dial, and an automatic **report-card** score. It is **not** a usable engine here (text-only, from-scratch GPT-2-class), but its five properties are exactly what make training agent-runnable, and we copy them: **(1)** one command, end-to-end; **(2)** one complexity dial (sane auto-derived defaults); **(3)** an automatic **report card** ? a deterministic pass/fail vs a target (the crux); **(4)** reproducible + deterministic + cheap; **(5)** minimal/readable.

Engine (don't reinvent): **`ms-swift`** already delivers this for VLMs ? CLI-driven, **native video**, Qwen3-VL support, an eval backend (EvalScope), and reproducibility via `args.json` + `full_determinism`. (LLaMA-Factory is the close second; HF **`nanoVLM`** ? ~750 lines ? is the minimal reference to *learn* VLM training, not the production path.)

The harness = thin glue over components we already have ? one declarative,

agent-invokable job (call it `train_speedrun`):
1. **Build dataset** from golden labels ? clip?QA pairs (reuse the `annotation` registry +
   `candidate_segments` + golden set ? the labeling seam in `backend/`).
2. **Fine-tune** via ms-swift on a **ComputeBackend** GPU (§3.2), run as an orchestration
   job (§5 / DEFERRED #16): one config, idempotent, logged.
3. **Auto-eval + report card** ? extend the existing `backend/eval/` regression harness to
   QA quality; emit a `report.md` pass/fail vs the **held-out golden QA eval**.
4. **Register** the result as the `owned_vlm` provider (Gemini paid fallback stays) **only
   if it beats the bar**.

So: **golden labels ? ms-swift train (on a compute backend) ? eval gate ? register
provider**, behind one command an agent or CI runs. nanochat is the inspiration for the
wrapper; ms-swift is the engine; the **golden QA eval is the report card**.

> **Prerequisite (the crux):** the harness is only as trustworthy as the **golden QA eval
> set**. nanochat's speedrun is meaningful *because* its eval (CORE score) is automatic and
> trustworthy; ours is the golden QA eval ? **build that first**, or an agent-run loop
> optimizes blind. (Same "moat = dataset + eval", scaffolding-first principle.)

?? Teacher / distillation constraint (the single biggest legal landmine)
You **cannot** train the owned model on outputs from the paid frontier APIs ? a general
video-QA assistant is squarely a *"competing model"*: **Gemini** ("may not use the
Services to develop models that compete"), **OpenAI** ("may not use Output to develop
models that compete"), **Anthropic** (bars using Outputs to train competing models). Two
safe lanes: **(1)** generate synthetic QA with **open-model teachers only** (Qwen-VL /
InternVL / Gemma 4 ? Apache-clean); **(2)** keep the paid API strictly as a **runtime
fallback** ? serving answers is fine, *harvesting its outputs to train the model that
replaces it is not*. Write a hard **"no training on competitor-API outputs"** rule into
eng + legal process. (This makes the ROADMAP's "Gemini as offline teacher only" precise:
the teacher is *open models*, not the paid APIs.)

Data flywheel, personalization & consent tiers
Two complementary loops, each with its **own** opt-in (training consent is separate from
processing consent ? §7):

- **Global model (shared ? raises the floor).** Train the base owned model on an
  explicitly-contributed, licensed corpus ? *not* on whatever sits in a tenant's bucket
  (preserves "your video, your storage"). Flywheel: *more consented, labeled matches ? a
  better shared model ? better highlights/answers ? more reason to contribute.* Higher
  consent bar (pooled data, biometric/minor care ? §7); **minors excluded by default**.

- **Per-user adapters (personalized ? raises the ceiling) ? owner hunch, under exploration.**
  A small **per-user (or per-tenant) LoRA adapter, fine-tuned *only on that user's own
  consented videos*** on top of the shared base, served via **multi-LoRA** (LoRAX /
  Together / Predibase). Why it's compelling:
  - **Precision:** adapts to *that user's* court, camera angle, lighting, and players'
    styles ? materially better detection/highlights for them.
  - **Cleanest consent story:** *"improve **your** detection using **your** data"* is far
    lower-friction and lower-risk than contributing to a global model ? the data improves
    only the user's own adapter, with **no cross-user mixing**, sidestepping much of the
    pooled-corpus biometric/minor entanglement. Reinforces the differentiator: *your video,
    your storage, your compute ? and now **your model**.* With BYO-compute the adapter can be

    trained/served on the user's own GPU and **never leave their boundary**; **deletion = drop the adapter** (clean erasure).
  - **Open questions (still a hunch):** cold-start (need enough of the user's labeled data before a personal adapter beats the base ? fall back to the shared model meanwhile); a **per-user held-out eval** to prove the adapter helps (don't ship a regression); base + optional global-adapter + per-user-adapter stacking; per-user training cost at scale. Worth a spike once the harness + golden eval exist.

Two consent tiers, then: **(a)** "improve **my own** detection with **my** data" (per-user, low friction) and **(b)** "contribute to the **shared** model" (global, higher bar). **Default for both is off.**

6. Phased roadmap

Phase	Deliverable	Build / reuse
P0	**Async job model, single-tenant** (DEFERRED #16)	`jobs` table via `user_version` ladder; worker; 202+status endpoints
P1	**Identity + multi-tenant data model + upload + highlight persistence + history**	auth (build vs Auth0/Clerk/Supabase ? decide); Postgres + `tenant_id`/RLS; `highlights` table; repository layer; history API + UI
P2	**`StorageBackend` registry + BYO-storage**	new ABC+registry; `local`?`gdrive`/`s3`/`gcs`; replace `video_path` with `StorageRef`; encrypted per-tenant creds
P3	**`ComputeBackend` registry + BYO-compute**	`local`/`hosted`/`byo_worker`; pull-based worker agent; control/data-plane split
P4	**Rich rally/shot semantic metadata & tagging** *(the next value stage after solid rally detection + highlights)*	shot type (smash / backhand drive / net), rally **outcome** (winner / forced / **unforced error**), context (e.g. **open court but error**) ? query-API filters + `HighlightMetadata` + a **human correction/review loop** feeding the golden set via the `annotation` registry + `candidate_segments`
P5	**Owned generative video-QA model** (§5A)	global flywheel **+ per-user adapters**; P4 metadata is training signal ? fine-tune an open VLM (Qwen-VL) via LoRA ? serve as `owned_vlm` **provider with Gemini paid fallback**; extend the eval harness to QA quality
P6 *(visionary)*	**Gait / biomechanics under a supervising coach**	deliberately **back-burner**; technique feedback; highest Art. 9 bar ? gate behind explicit biometric + parental consent (§7)

P0?P1 deliver the headline ask (per-user upload + stored highlights + history). P2?P3 deliver the "plug-and-play storage/compute" ask. **P4 (rich semantic metadata) is the immediate next value stage once rally detection + highlights are solid** ? e.g. *"rallies that ended in a smash or backhand drive,"* *"rallies where the court was open but the player made an unforced error."* **P5** owns the model; **P6 (gait) is the visionary horizon, intentionally deferred.**
Sequencing principle (owner directive): make the scaffolding (P0?P4) **rock-solid first**, keep the **known paid stack as an always-ready fallback**, and only then shift focus to **owned-model quality** (P5). The provider registry keeps the paid path one config flip away.

7. Privacy & legal (gait raises the bar ? design for it now)

- **The load-bearing nuance:** biometric data is special-category **only when used to *uniquely identify* a person** (GDPR Art. 9). **Highlight/shot detection on players a coach already knows is general analytics ? generally *not* Art. 9.** But **gait/biomechanics that builds a per-athlete identifying template tips into Art. 9** and needs **explicit, granular consent** referencing biometric data. Engineer gait with that boundary (consent gate, purpose limitation, no cross-account identification);

decide whether it runs **locally/on-tenant only** (strong posture) vs our cloud. (EDPB Guidelines 3/2019 on video devices apply.)
- **Minors (e.g. academies).** Parental/guardian consent and safeguarding are first-order, not an afterthought. Model consent explicitly per relationship (user?subject, coach?athlete where the overlay applies, parent?minor).
- **BYO-storage is the central privacy/liability lever:** if raw video stays in the tenant's own bucket/Drive and we process via signed URL or their compute, we minimize what we hold ? shrinking breach surface, residency, and erasure obligations.
- **Paid-LLM no-train (C4):** use **Gemini via Vertex AI** (contractual no-training), **not the consumer/free tier**; set short/zero retention, document it in the DPA, and surface "your footage is never used to train AI" as a trust feature for academies. For the gait phase, prefer keeping biometric inference off third-party APIs entirely.
- **Training consent is a distinct, higher bar (for the owned model, §5A).** Training on user video needs a **separate, explicit, opt-in training licence** ? not bundled with processing consent ? and it **compounds** the biometric (Art. 9) and minor-consent issues above. Train **only on explicitly contributed/licensed video**; never train on what merely passes through a tenant's bucket. **Exclude minors' video from the training pool by default**, keep the training corpus a **separate, firewalled, opt-in sub-pool**, and log **per-clip consent/provenance** for revocation + audit. Incentives must not make consent non-free (the product stays fully usable without contributing). And the **teacher-distillation rule** (§5A): never train on paid-LLM outputs. **Two consent tiers (§5A):** *(a)* "improve **my own** detection with **my** data" ? a per-user adapter trained only on the user's own video (lowest friction; no cross-user mixing; deletion = drop the adapter); *(b)* "contribute to the **shared** model" ? pooled corpus, higher bar. Default for both is **off**.
- **Right to erasure / retention:** per-tenant, per-athlete delete must cascade across DB + storage artifacts (the `ON DELETE CASCADE` discipline already in `database.py` extends here).

8. Key decisions for the owner (the forks)

1. **Hosted-first vs BYO-first.** Recommended: **hosted control-plane first** (P0?P1) with **BYO-storage soon after** (P2); **BYO-compute later** (P3) as a power-user/enterprise option ? it's the highest-complexity, highest-security item.
2. **Database now or later.** Move to **Postgres at P1** for the hosted path, or stay SQLite-per-tenant if self-hosted-single-academy is the first GTM. (Abstract DB access regardless.)
3. **Auth: build vs buy** (Auth0 / Clerk / Supabase Auth / Cognito).
4. **Orchestration tech** (see research recommendation) ? bias to the simplest that gives retries + status.
5. **Gait processing location** ? local/on-tenant (privacy-max) vs cloud (capability-max).
6. **How far to push BYO-compute** given its security/ops complexity.
7. **Owned-model base + licensing (§5A).** Which commercial-OK video VLM to build on (Qwen-VL Apache-2.0 vs Gemma 3 custom license vs ?) and confirm a gated commercial derivative is permitted ? see COMMERCIALIZATION C14.
8. **Training-data sourcing & consent.** Opt-in contribution terms, incentives, and the teacher-distillation constraint (whether paid-LLM outputs may seed the corpus).
9. **When to start the owned model** ? strictly after the scaffolding + flywheel are solid (the explicit directive), with the paid provider as the standing fallback.

Risks: scope explosion (sequence strictly P0?P5); multi-tenant data-leak bugs (RLS +

tests); BYO-compute security; **biometric/minor liability** (get this right before any gait feature ships); **training on user video without proper opt-in/licence**, and **base-model / teacher-distillation licence violations**; cost of hosted GPU (training *and* inference); **premature model-building** before data/eval are ready (mitigated by keeping the paid fallback and gating P5 behind the flywheel).

9. Sources

External best-practices research (2026-06-07) backing the recommendations above. Each underlying claim in the research was tagged fact-vs-synthesis; figures are as-of and should be re-validated before committing budget.

Storage abstraction: [fsspec](https://fsspec.com/) · [cloudpathlib](https://github.com/drivendataorg/cloudpathlib) · [smart_open](https://github.com/piskvorky/smart_open) · [Apache Libcloud](https://libcloud.apache.org/) · [gdrive-fsspec](https://github.com/fsspec/gdrive-fsspec) · [dropboxdrivefs](https://github.com/fsspec/dropboxdrivefs) · [AWS S3 presigned URLs](https://docs.aws.amazon.com/AmazonS3/latest/userguide/using-presigned-url.html) · [token-based S3 with STS](https://oneuptime.com/blog/post/2026-02-12-token-based-access-s3-sts/view) · [signed URLs across clouds](https://medium.com/@shaikhsarwan49/signed-urls-in-aws-gcp-and-azure-a-beginner-friendly-guide-e2e549425865)

Multi-tenancy: [Bytebase: multi-tenant DB patterns](https://www.bytebase.com/blog/multi-tenant-database-architecture-patterns-explained/) · [Hunchbite: RLS vs schema-per-tenant](https://hunchbite.com/guides/multi-tenant-saas-architecture) · [PlanetScale: tenancy in Postgres](https://planetscale.com/blog/approaches-to-tenancy-in-postgres) · [DEV: DB-per-tenant vs shared schema](https://dev.to/young_gao/multi-tenant-architecture-database-per-tenant-vs-shared-schema-1n2e)

BYO-compute / orchestration: [Pinecone BYOC](https://www.pinecone.io/blog/byoc/) · [Nuon: control/data-plane](https://nuon.co/blog/byoc-control-plane-data-plane-architectures) · [GitHub self-hosted runners (ARC)](https://docs.github.com/en/actions/hosting-your-own-runners/managing-self-hosted-runners-with-actions-runner-controller/about-actions-runner-controller) · [SkyPilot](https://github.com/skypilot-org/skypilot) · [SkyPilot AI control plane](https://blog.skypilot.co/ai-job-orchestration-pt2-ai-control-plane/) · [Modal (AWS Marketplace)](https://aws.amazon.com/marketplace/pp/prodview-j727623xqhh2k) · [Northflank: Anyscale alternatives](https://northflank.com/blog/anyscale-alternatives-for-ai-ml-model-deployment) · [DevPro: Python task queues 2025](https://devproportal.com/languages/python/python-background-tasks-celery-rq-dramatiq-comparison-2025/) · [Judoscale: choosing a task queue](https://judoscale.com/blog/choose-python-task-queue) · [Hatchet vs Temporal](https://hatchet.run/versus/hatchet-vs-temporal) · [Hatchet](https://github.com/hatchet-dev/hatchet)

Privacy / legal: [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [GDPR Local: biometric compliance](https://gdprlocal.com/biometric-data-gdpr-compliance-made-simple/) · [Legiscope: Art. 9](https://www.legiscope.com/blog/gdpr-article-9-special-categories.html) · [EDPB Guidelines 3/2019 (video)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_201903_video_devices_en_0.pdf) · [Fox Rothschild: EDPB video guidance](https://www.foxrothschild.com/publications/video-surveillance-under-gdpr-edpb-issues-final-guidance) · [ICO: children & UK GDPR](https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/childrens-information/children-and-the-uk-gdpr-old/what-are-the-rules-about-an-iss-and-consent/) · [EDPB Guidelines 05/2020 (consent)](https://www.edpb.europa.eu/sites/default/files/files/file1/edpb_guidelines_202005_consent_en.pdf) · [AI data-retention compared](https://ax-sentinel.com/blog/ai-data-retention-policies-compared) · [Xenoss: enterprise LLM platforms](https://xenoss.io/blog/openai-vs-anthropic-vs-google-gemini-enterprise-llm-platform-guide)

Comparable products: [Veo](https://www.veo.com/) · [Veo Help Center](https://support.veo.com/hc/en-us/articles/4459159526929-Overview-of-cam-veo-co-Record-manage-and-upload-footage-with-your-Veo-Cam-1) · [Latterly: Hudl competitors](https://www.latterly.org/hudl-competitors/) · [Apple: Behind the Design ? SwingVision](https://developer.apple.com/news/?id=0pg4dthn) · [SwingVision FAQ](https://swing.vision/faq) · [Sportsvisio: coach video-analysis comparison](https://www.sportsvisio.com/stories/video-analysis-software-for-coaches-complete-comparison)

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561 **Training backends & harness (§5A):** [karpathy/nanochat (harness reference)](https://github.com/karpathy/nanochat) · [HF nanoVLM (minimal VLM)](https://github.com/huggingface/nanoVLM) · [nanoVLM blog](https://huggingface.co/blog/nanovlm) · [Tinker (GA + vision)](https://thinkingmachines.ai/blog/tinker-general-availability/) · [Tinker model lineup](https://tinker-docs.thinkingmachines.ai/model-lineup) · [Tinker ToS (weights/no-train)](https://thinkingmachines.ai/legal/tos/) · [Fireworks VLM fine-tuning](https://fireworks.ai/blog/vlm-tuning) · [Together AI fine-tuning](https://www.together.ai/fine-tuning) · [Predibase: fine-tune & serve VLMs](https://predibase.com/blog/how-to-fine-tune-and-serve-vlms-in-predibase) · [Predibase RFT/GRPO](https://docs.predibase.com/user-guide/fine-tuning/grpo) · [ms-swift (native video/audio)](https://github.com/modelscope/ms-swift) · [LLaMA-Factory (VLMs)](https://github.com/hiyouga/LlamaFactory) · [Unsloth (Qwen3-VL)](https://qwen.readthedocs.io/en/latest/training/unsloth.html) · [Fine-tuning tools 2026 (LoRA/QLoRA)](https://www.index.dev/blog/top-ai-fine-tuning-tools-lora-vs-qlora-vs-full)
562
563 **Owned model & licensing (§5A):** [Qwen2.5-VL](https://qwenlm.github.io/blog/qwen2.5-vl/) · [Qwen2.5 license split](https://qwenlm.github.io/blog/qwen2.5/) · [Qwen3-VL](https://github.com/QwenLM/Qwen3-VL) · [Gemma 3 (multimodal)](https://huggingface.co/blog/gemma3) · [Gemma Terms of Use](https://ai.google.dev/gemma/terms) · [Gemma license-risk analysis](https://wcr.legal/google-gemma-license-risks/) · [Gemma 4 / Apache-2.0](https://www.mindstudio.ai/blog/what-is-gemma-4-apache-2-license-commercial-ai-deployment) · [Llama 3.2 license (700M MAU + naming)](https://www.llama.com/llama3_2/license/) · [gpt-oss (text-only, Apache-2.0)](https://openai.com/index/introducing-gpt-oss/) · [Gemini API terms (compete clause)](https://ai.google.dev/gemini-api/terms) · [OpenAI terms](https://openai.com/policies/row-terms-of-use/) · [Anthropic: outputs & training](https://support.claude.com/en/articles/12326764-can-i-use-my-outputs-to-train-an-ai-model) · [QLoRA practical](https://apxml.com/courses/fine-tuning-adapting-large-language-models/chapter-4-parameter-efficient-fine-tuning/practice-qlora-fine-tuning) · [VLM LoRA cookbook](https://huggingface.co/learn/cookbook/en/fine_tuning_vlm_trl) · [GDPR Art. 9](https://gdpr-info.eu/art-9-gdpr/) · [athletes & biometric data](https://www.architectureandgovernance.com/applications-technology/biometric-data-and-athletes-privacy-law-and-compliance-implications/) · [federated learning + DP](https://research.google/blog/federated-learning-with-formal-differential-privacy-guarantees/) · [self-hosted LLM VRAM/cost](https://www.gpu-mart.com/guides/self-hosted-llm) · [Gemini API pricing](https://ai.google.dev/gemini-api/docs/pricing)
564
565 ## See also
566 - [`COMMERCIALIZATION.md`](COMMERCIALIZATION.md) ? C12 (SaaS foundations), C4 (paid-LLM), economics.
567 - [`DEFERRED_HARDENING_PLAN.md`](DEFERRED_HARDENING_PLAN.md) ? #16 async job model (the P0 infra slice), #7 packaging.
568 - [`PMF_MARKET_RESEARCH.md`](PMF_MARKET_RESEARCH.md) ? academy/coach beachhead (who this serves).
569 - [`CODE_MAP.md`](CODE_MAP.md) ? current architecture + the registry extension points.
570 - [`ROADMAP.md`](ROADMAP.md) ? offline self-improving rally detection (the pipeline's own trajectory).
571

5. assistant (2026-06-09T11:53:46.920Z)

Let me read the two owned-model docs that are the direct subject of these questions, plus the training study, to ground my recommendations in the repo's existing plan.

6. user (2026-06-09T11:53:47.898Z)

1 # Owned-model implementation plan (DRAFT ? for owner review, no code started)
2
3 **Status:** PLAN ONLY. No owned-model implementation begins until the owner greenlights a specific
4 milestone (per CLAUDE.md). Paid Gemini stays the always-ready primary/fallback throughout.
5 **Grounded in:** `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` (the verified

study) +
`docs/archives/quality-iterations/2026-06-multivideo-lovo/` (the LOVO + prototype that de-risk it).

North star & non-negotiables
- **Goal:** an OWNED, on-device rally?highlight model; local inference = the privacy moat. The moat is the
  consented dataset + eval harness, **not** the weights.
- **Guardrails (hard):** Apache-2.0/MIT bases only (Gemma-3 rejected); **never** train on paid-LLM
  (Gemini/OpenAI/Anthropic) outputs; exclude minors + separate training opt-in; split data **by match**.
- **Sequencing discipline:** committed next platform slice is the **async job model (scaffolding)**. Heavy
  model *training* (Gen-2, cloud) waits for P0?P4 scaffolding + a healthy corpus + explicit go. The
  corpus + data + compliance foundation is no-regret and can proceed first ? it's what makes later
  training cheap and clean.

Why now is the right moment (today's evidence)
- Heuristic cross-video LOVO = **0.443 ± 0.279** (Adarsh 0.722, Kushagra 0.163-ceiling). The ±0.279 is
  method variance ? the unsupervised windowing is footage-fragile even when WASB recall is excellent
  (Kushagra: 97% of rallies have shuttle motion, yet unsegmentable).
- The learned prototype gets **held-out per-frame AUC 0.84?0.88**; Kushagra oracle F1 **0.578 vs 0.163**.
  ? The rally signal is *learnable*; the bottleneck is **data scale + calibration (n?5), not method**.

Phase 0 ? Foundation (start now; no GPU/scaffolding dependency)

M0.1 ? The labeled-corpus spine (the moat) · *owner-greenlight item*
- **What:** one canonical JSONL row per rally: `{video_id (file MD5), rally_ordinal, sport, start, end,
  ending_reason, label_type, source(human|pseudo|vendor), annotator/provenance, consent/training_opt_in,
  split(train|val|test, assigned BY MATCH), trajectory_ref, labeled_window}`.
- **Where:** authored in `sports-data-collector` (system-of-record; extends PR #13 + the positioning doc);
  the indexer only consumes transcoder outputs.
- **3 deterministic transcoders (one row ? three views, zero re-labeling):** (1) heuristic/LogReg
  candidate labels (already exists: `training.label_candidates`), (2) TAL ActivityNet-JSON, (3) VLM clip?QA.
- **Guardrail enforcement:** provenance/consent/minors fields gate corpus membership *before* transcoding;
  Gemini labels never enter a training transcoder; the new `eval_partial_labels` window flows into every view.
- **One scorer spine:** `metrics.match_intervals` for heuristic-labeling, TAL eval, and the Gen-2 reward.
- **Acceptance:** round-trip test (collector export ? each transcoder ? indexer loaders); split-by-match
  enforced; no Gemini-sourced rows in any training view.

M0.2 ? Grow n?2 ? 5 distinct matches · *owner action + my scoring loop*
- **What:** tag ?3 more *distinct* matches (gBaaddy, TestLarge, + a small Roora batch). I generate the
  trajectory + score each as it lands (the loop we've been running).
- **"Healthy corpus" bar:** ?5 distinct match-sessions with full human labels ? stable leave-one-match-out.
- **Acceptance:** n?5; per-match + LOVO numbers recorded; corpus diversity

(camera/court/lighting) noted.

M0.3 ? Compliance cleanup (do before it bakes in) · *owner-greenlight item*
- **(i) Purge Gemini-derived TRAINING labels.** The distilled pipeline currently trains on Gemini silver (`training.label_candidates` over Gemini CSVs) ? this **violates the no-paid-LLM-training rule.** Retarget training to human + open-teacher (WASB/own-model) labels only; Gemini stays eval/teacher-reference.
- **(ii) WASB checkpoint provenance (C3).** The distributed `wasb_badminton_best.pth.tar` is academic-data-trained (provenance unresolved) ? it propagates to anything trained on its features. **Decide:** document as a hard Gen-0 prerequisite and plan ROADMAP Phase-4 "train own WASB weights," or scope an alternative.
- **Acceptance:** no paid-LLM outputs in any training path; a written C3 decision + prerequisite flag.

M0.4 ? Gen-0 A/B harness (built here; trains on your GPU box) · *owner-greenlight item*
- **What:** productionize the prototype into a real harness ? feature stack (visibility-run-length, velocity, acceleration, net-crossing flags, smoothing/windowing) ? **ActionFormer (MIT, 1:1 rally=instance)** and/or **ASFormer (MIT, TAS)** train?eval?**ship-gate**. Reuse `metrics.match_intervals`.
- **Ship gate (the honest bar):** held-out per-match F1 must beat the **honest cross-video heuristic LOVO (~0.44)**, not the fit-on-self 0.73. nanochat-style pass/fail verdict.
- **Where it runs:** training/eval on the **GPU/WSL machine** (can't run in this dev container). Harness + feature stack are authored here; you run the A/B.
- **Acceptance:** one-command train?eval?verdict; reproducible; the prototype's AUC translated to segment F1 at n?5; documented A/B (ActionFormer vs ASFormer vs heuristic).

Phase 1 ? Gen-0 ship decision (gated on n?5 + M0.4)
- Run the A/B; if a learned Gen-0 clears the honest LOVO bar with per-video calibration, **it becomes the rally segmenter** (still fully local, MIT, on the 2070 Super). If not, the finding (more data / better features needed) is itself the next-step signal ? no over-fitting to declare victory.
- **Decision gate:** Gen-0 adopted ? more data/features first. Either way, the corpus + harness persist.

Phase 2 ? Gen-1 / Gen-2 (gated on platform scaffolding P0?P4 + explicit go)
- **Gen-1 (Option A):** concatenate fully-local cue channels (audio-onset, pose/court) onto the same head; fusion in our rally layer. License/cost-vet the pose extractor (open item).
- **Gen-2 (Option B):** e2e multimodal VLM on an Apache-2.0 base (**Qwen3-VL-4B** primary / Qwen2.5-VL-7B fallback / **Gemma 4** end-state once a timestamp recipe is proven), SFT cold-start ? GRPO/RFT with a temporal-IoU-vs-golden reward. **Cloud training on Modal** (best no-access contract); base license re-verified per-checkpoint.
- **Hard pre-serving gate:** the **$/call crossover benchmark** (self-hosted-VLM vs Gemini on real workload) + an **on-device latency/throughput** measurement of the quantized base on the 2070 Super ? both currently *unmeasured* and load-bearing. Don't stand up an owned serving stack until these clear (COMMERCIALIZATION C7).

Decisions I need from you (to unblock specific milestones)
1. **Greenlight which Phase-0 milestone(s) to start in code** (M0.1 corpus spine = my recommendation; M0.3
   cleanup; M0.4 harness). M0.2 (tagging) runs regardless.
2. **WASB C3 stance** ? accept "document as prerequisite" for now, or prioritize own-WASB-weights sooner?
3. **Ratify "Apache-2.0/MIT"** (the study's approved list includes MIT bases/teachers; the guardrail text
   says "Apache-2.0 only").
4. **Corpus home** ? confirm the collector is the system-of-record for the spine (consistent with PR #13).

Open prerequisites (carried from the study's critic ? none block Phase 0)
WASB C3 provenance · Gemma-4 license stays AMBER until a raw HF LICENSE check · on-device VLM latency
unbenchmarked · real VLM VRAM/token cost on our 1080p/4K-60fps clips · DPA + no-train for any cloud training ·
the owned-vs-Gemini $/call crossover volume.

This plan is design-only and reversible. Nothing here starts until you greenlight a milestone.

7. user (2026-06-09T11:53:48.488Z)

Owned-model training study (rally segmentation ? end-to-end multimodal)

Date: 2026-06-08 · **STATUS: DESIGN BLUEPRINT (no implementation until PLATFORM P0?P4 + healthy corpus; owner sign-off required).**

> How to train our **OWNED** rally/highlight model **reusing the videos we label**, aligned to the
> Apache-2.0 / no-paid-LLM-training / privacy-local guardrails. Produced by an 11-agent research
> workflow (ground ? 5 web-research dimensions ? adversarial verification of costs/licenses/privacy ?
> synthesis + completeness critic; 35 sources). Empirical results from THIS session (LOVO + a learned
> prototype) are folded in and **resolve the study's biggest open unknown** ? see §"What we proved today".
> Numbers from web sources are 2026 snapshots in a ±15?25% band; **re-quote live before budgeting.**

Thesis

Train the owned model in **three staged generations off ONE model-agnostic labeled corpus** ? a label
collected once feeds every generation, so labels are never re-collected. **The moat is the consented
dataset + the eval harness, never the weights.**

Gen	When	Approach	Base / model	Compute
Gen-0	now ? unblocks at n?5 matches	tiny **temporal action segmentation (TAS)** head over a hand-built per-frame feature stack from the WASB trajectory we already produce; per-frame rally(1)/non-rally(0) ? merge into [start,end]	**ASFormer (MIT, ~1.1M params)** primary; **ActionFormer (MIT)** A/B challenger (overtakes once corpus = hundreds of matches)	**local RTX 2070 Super**, trains in minutes, $0
Gen-1 / Option A	6?12 mo	concatenate more local cue channels onto the SAME head: trajectory + audio-onset + pose/court serve-gating; fusion in our rally layer, cues swappable	same ASFormer/ActionFormer, wider input	local + occasional cloud burst (pseudo-labeling)
Gen-2 / Option B	12?24+ mo, after scaffolding + healthy corpus	end-to-end

multimodal VLM: (frames+audio+pose+trajectory) ? rally [start,end] (+semantics); SFT cold-start then **GRPO/RFT with a temporal-IoU-vs-golden reward** | **Qwen3-VL-4B-Instruct** (Apache-2.0, on-device) primary; **Qwen2.5-VL-7B** fallback; **Gemma 4 E4B/12B-Unified** (Apache-2.0 + native video+audio) end-state | rented 24?48 GB A100/L40S (does NOT fit 8 GB box) |

What we proved today (de-risks the blueprint)

The study's **"dominant empirical unknown"** was: *can a low-dim WASB trajectory (?3?4 raw dims) feed a TAS head, given ASFormer's small-data result used 2048-d I3D features?* This session answered it.

- **Cross-video LOVO of the current heuristic** (`calibrate_local`, Adarsh-96 + Kushagra-32, human labels): mean held-out F1 **0.443 ± 0.279**; Adarsh 0.722, **Kushagra 0.163** (its own ceiling ? *no* velocity-windowing config does better). **The ±0.279 spread is method variance, not config.** ? This also **corrects the study's "~0.73 bar"**: that was a fit-on-self single-match number; the honest cross-video heuristic bar is **~0.44**, which *lowers* the Gen-0 ship gate.
- **A learned per-frame prototype** (`backend/eval/rally_seq_proto.py`: 6 hand-built features ? visibility rate, mean/max shuttle speed, **net-crossing rate**, x/y spread ? into the repo's dependency-free LogisticRegression, honest LOVO): **per-frame AUC 0.878 (Adarsh) / 0.841 (Kushagra), held-out.** On Kushagra ? the match the heuristic *cannot* segment (0.163) ? a calibrated threshold reaches **segment F1 0.578 (3.5×)**. The naive cross-video threshold gives only 0.049, so **the remaining gap is threshold/operating-point calibration at n=2, not representation.**

- **n=3 update (doubles match added ? `LargeTestVideo`, 49 rallies):** the learned prototype **overtakes the heuristic.** Mean held-out LOVO F1 ? learned **0.377 ? 0.615**, heuristic **0.443 ? 0.544**. Kushagra's learned held-out F1 jumped **0.049 ? 0.417** (one added match closed most of the calibration gap); the **doubles** match, held out and trained only on the two singles, scored AUC **0.914** / F1 0.729 (generalizes across play type + frame rate). The heuristic still cannot rescue Kushagra at any n (0.151, capped). **The honest "bar to beat" is thus ~0.54 (n=3), not 0.73 ? and the learned path already clears it on the mean and keeps rising with n.**

Conclusion: the rally signal IS learnable from the WASB trajectory features, cross-footage (AUC ~0.85?0.91) ? the heuristic just can't exploit it. This **validates the Gen-0 ASFormer-over-WASB-features path**, shows the hand-built feature stack the study calls for is the right lever, and confirms the bottleneck is **data + per-video calibration (n?5 matches), not method.** The prototype is the seed of Gen-0, and the n=2?n=3 trend (learned mean +0.24, overtaking the heuristic) is the empirical green light.

Platform ranking (privacy + cost lens; the fine-tune is a sub-24h, $10?50 job ? privacy/ergonomics dominate)

1. **Modal ? PRIMARY.** Strongest *written* no-access contract (verbatim: "Modal will never access? the inputs/outputs to your Functions, or any data you store"), ?7-day TTL, SOC2-II, HIPAA-BAA; per-second scale-to-zero matches retrain-on-each-batch. A100 80 GB ~$2.50/hr, H100 ~$3.95/hr (corrected down from cited). Privacy is the load-bearing reason, not price.
2. **RunPod (Secure Cloud tier ONLY) ? FALLBACK / Option-B scale-up.** Cheapest vetted-

DC self-serve; A100 80 GB ~$1.39?1.49/hr. Community/P2P tier is UNVETTED ? **never** touches consented video. Confirm AES-256/DPA at booking.
3. **Vertex AI / GCP ? ON BENCH.** Only strong-privacy option with an India region, but it's **capacity-gated** (asia-south1 high-end GPU is invitation-only, H100-class A3 not A100) and **tuning Gemini yields NO portable on-device weights** ? only tuning an Apache-2.0 base in Model Garden does. Use only if India residency-at-rest is mandated.
4. **Self-hosted RTX 2070 Super 8 GB** ? Gen-0 training + ALL inference + CI only. **Cannot** QLoRA a 7B video VLM (~20?24 GB). Best privacy (nothing leaves the box).
5. **Fireworks** (managed, contractual no-train + ZDR) usable; **Together BLOCKED** until a signed DPA + explicit no-train clause. AWS SageMaker / Azure ML = heavy for a solo founder; India-residency only.

Label reuse ? one corpus, three transcoders (the actual moat)

Promote the rally CSVs from eval-only files to **one model-agnostic corpus**: JSONL, one row per
`(video_id, rally_ordinal)` = `{start, end, ending_reason, sport, source (human|pseudo|vendor),
annotator/provenance, consent/opt-in, label_type, split (BY MATCH not clip), trajectory_ref}`. The
provenance/consent fields are the **enforcement point** for the legal guardrails. Three deterministic
transcoders read it, zero re-labeling:
1. **Heuristic/LogReg view** ? already exists (`backend/eval/training.py::label_candidates` via `metrics.match_intervals`).
2. **TAL view** ? ActivityNet JSON (`segment:[start,end]`, `label:'rally'`); our seconds map 1:1.
3. **VLM clip?QA view** ? "list every rally [start,end] in this clip"; `ending_reason` ? a second QA task.

One scorer spine: reuse `metrics.match_intervals` for heuristic-labeling, TAL eval, AND the Gen-2 RFT
reward ? no metric drift. **Leakage guard:** split **by match, never by clip**. **Amplification (all
guardrail-clean):** open-teacher pseudo-labels (WASB/own-model, confidence-gated ? never paid-LLM),
boundary-preserving temporal augmentation, active learning to spend the Roora budget on highest-uncertainty
matches. ? **The new partial-label windowing** (`eval_partial_labels.restrict_to_window`) must flow into the
TAL/VLM views so prefix-labeled videos aren't mis-scored (a corpus that's partially tagged is the norm).

Licensing & privacy guardrails (all respected; verify per-checkpoint)

- **Bases: Apache-2.0/MIT only.** APPROVED: Qwen2.5-VL-7B/32B (Apache-2.0), Qwen3-VL-4B/8B (Apache-2.0),
  Gemma 4 E2B/E4B/12B-Unified (Apache-2.0 + native video/audio), InternVL3 ?38B (MIT/Apache), ASFormer/
  ActionFormer/TriDet (MIT). **REJECTED:** Gemma 3 (viral Gemma Terms + PUP), Qwen2.5-VL-3B (NC research),
  Qwen-72B/InternVL3-78B (100M-MAU cap), Llama Vision (700M-MAU), DAMO VideoLLaMA3 weights (NC + paid-LLM-trained),
  TimeChat/VTimeLLM/Momentor (NC/no-license = method-reference only), VideoMAE weights (CC-BY-NC), `yabufarha/ms-tcn`
  (Commons Clause ? use `sj-li/MS-TCN2` MIT). ? **Gemma 4 size names = E2B/E4B/26B-A4B/31B/12B-Unified ? there is no plain "4B".**
- **No paid-LLM training data** ? Gemini/OpenAI/Anthropic = inference/eval/teacher-reference only, **never** a
  training signal. Gemini silver labels stay out of every transcoder.
- **Exclude minors by default + separate training opt-in;** per-user adapters trained only on that user's
  consented video (deletion = drop the adapter). Enforced at the corpus-membership layer before transcoding.
- **Privacy lever = local inference** (Gen-0 ~1.1M params; Gen-2 4B quantized). Any

cloud *training* needs a signed DPA + no-train assurance; route only owned/consented footage, never user uploads.

Cost (spend on labels, not GPUs)

- **Gen-0:** ~$0 (local 2070 Super; trains in minutes; ~4.5 GB VRAM).
- **Gen-2 LoRA:** ~$5?40/run on a rented 24?48 GB GPU (RunPod cheapest; **VRAM/token cost on real 1080p/60fps clips is UNVERIFIED ? 24 GB is a floor, re-benchmark**). Managed (Fireworks/Together) ~$2?30 *if* text-priced (video tokens can balloon). RFT = compute-heavy (multiple rollouts/sample).
- **Inference crossover (the unanchored number):** self-hosted 7B beats Gemini (~$0.05/10-min clip) **only** above sustained QA volume; below it, **paid Gemini is cheaper** ? paid-primary now, owned-primary later. **Benchmark $/call before committing** (COMMERCIALIZATION C7).

Open risks & critic caveats (read before acting)

- **?? WASB checkpoint provenance (C3) propagates.** The study calls WASB "MIT," but that's the *code*; the distributed `wasb_badminton_best.pth.tar` is **academic-data-trained, provenance unresolved** (COMMERCIALIZATION C3, still open). Every Gen-0/1 plan uses WASB features as input AND as a pseudo-label teacher ? a "clean" owned model trained on provenance-unclear features **does not launder the provenance.** ROADMAP Phase-4 "train own WASB weights" is a **prerequisite, not an aside.**
- **Gemma 4 "clean + native-video" rests on post-cutoff secondary sources** ? keep AMBER until a raw HF LICENSE revision is verified, given the Gemma-3 rejection. Its timestamp-emission recipe is less documented than Qwen's.
- **Existing distilled model was trained on Gemini silver labels** ? the (correct) eval-only policy now **forbids this; that pipeline must be purged of Gemini-derived training labels** to comply.
- **Policy ratifications needed:** guardrail literally says "Apache-2.0 only" but the approved list includes MIT ? ratify "Apache-2.0/MIT". The MAU-cap rejections (72B/78B) may be over-conservative at our scale.
- **On-device privacy claim is unbenchmarked** ? no tokens/sec for a quantized 4B VLM on 8 GB Turing at 1080p/60fps. **Measure before relying on it.**
- **Amplification is load-bearing for Gen-2** ? 128 raw rallies can't reach the ~5k?20k clip?QA band alone; if augmentation + QA-multiplexing + pseudo-labels underdeliver, Option B is under-fed. Amateur 60fps may need more labels than broadcast-trained literature implies.
- **Dropped option worth keeping:** InternVideo2 encoder (MIT) + light grounding head ? the *only* data-realistic owned-video experiment available at n?128 today. And: a **one-time ~$1?2k 24 GB local card** (e.g. used 3090) would preserve the privacy lever for *training* and dissolve most of the cloud-DPA analysis ? uncosted here.

Open questions (need the GPU/WSL machine-side suite ? not runnable in this dev container)

1. Does ASFormer/ActionFormer over the WASB feature stack beat the **honest ~0.44 LOVO** bar at n?5 matches?
 (Today's prototype says the per-frame AUC is there ? 0.84 ? so this is now *likely*; confirm at segment level.)
2. Real VRAM + token cost of fine-tuning Qwen3-VL-4B / Qwen2.5-VL-7B on our actual clip?QA frame counts?
3. Working Gemma-4 timestamp-emission recipe + exact checkpoint (E4B vs 12B-Unified)?

```
133     4. Measured self-hosted-7B $/call + latency vs Gemini on the target workload (the owned-
model ROI crossover)?
134     5. Will Together/Lambda/Azure sign a DPA with explicit no-train for consented video,
India-configurable?
135     6. Realistic clip?QA yield from 128 rallies via augmentation + QA-multiplexing + pseudo-
labels?
136
137     ## Sequencing (owner directive, non-negotiable)
138
139     This is a **design blueprint only.** It unblocks PLATFORM **P5** but starts **no
implementation** until P0?P4
140     scaffolding (async jobs, storage/compute backends) is rock-solid AND the golden corpus
is healthy. Paid Gemini
141     stays the always-ready primary/fallback. The only items safe to act on now ? **and even
these need owner
142     sign-off before code** ? are (a) the model-agnostic corpus format and (b) growing n?2 ?
n?5 matches (a small
143     Roora batch), which unblocks honest tuning immediately and is the **highest-ROI spend
(labels, not GPUs).**
144
145     *Full agent transcript + 35 sources: workflow `wf_49aab888-e10` (this session).*
146
```

8. assistant (2026-06-09T11:53:54.376Z)

I now have full grounding. The repo's existing blueprint already lands close to the research, with a few key corrections needed (on-device reframe, multisport from day one, RacketVision refutation, Qwen3-VL dating). Let me verify two repo facts that the research leans on: the `metrics.match_intervals` scorer and the existing `events`/`play_segments` schema that the bookkeeping recommendation says "mostly exists."

9. user (2026-06-09T11:53:56.586Z)

```
Found 21 files
backend\eval\rally_seq_proto.py
backend\reporting\harvest.py
backend\reporting\spec.yaml
backend\tools\export.py
backend\eval\batch.py
backend\main.py
backend\pipeline\segmenters\base.py
backend\pipeline\segmenters\motion.py
backend\reporting\__init__.py
backend\pipeline\validators\base.py
backend\pipeline\segmenters\yolo_hybrid.py
backend\pipeline\segmenters\gemini.py
backend\infrastructure\database.py
backend\eval\harness.py
backend\eval\distill_local.py
backend\api\static\app.js
backend\annotations\base.py
backend\sports\__init__.py
backend\providers\__init__.py
backend\eval\training.py
backend\eval\metrics.py
```

10. user (2026-06-09T11:53:57.510Z)

```
28:                CREATE TABLE IF NOT EXISTS videos (
39:                CREATE TABLE IF NOT EXISTS play_segments (
54:                CREATE TABLE IF NOT EXISTS events (
65:            # Versioned migrations live here. PRAGMA user_version is the schema
68:            version = cursor.execute("PRAGMA user_version").fetchone()[0]
75:                    CREATE TABLE IF NOT EXISTS candidate_segments (
```

```
            cursor.execute("PRAGMA user_version = 1")
        # future: if version < 2: cursor.execute("ALTER TABLE ..."); PRAGMA user_version = 2
```

11. assistant (2026-06-09T11:54:02.394Z)

The repo facts check out: `metrics.match_intervals` exists as the scorer spine, the `play_segments` / `events` / `candidate_segments` tables exist with a `user_version` migration ladder, and `rally_seq_proto.py` is the de-risking prototype. The conversational-QA "event store" the research recommends largely exists already. I have everything I need to synthesize.

12. user (2026-06-09T11:56:13.092Z)

Structured output provided successfully

13. assistant (2026-06-09T11:56:26.956Z)

I've completed the analysis and returned the structured recommendation. The answer integrates the repo's existing blueprint (which I verified against PLATFORM_ARCHITECTURE.md, OWNED_MODEL_IMPLEMENTATION_PLAN.md, the training study, and the actual database/metrics/prototype code) with the research and the verification corrections.

Key decisions delivered across the four questions:

- **Q1 (model + starting point):** Clarified the category confusion explicitly ? ms-swift is a *framework/forge*, Gemma 4/Qwen3-VL are *bases*, two orthogonal picks. Recommended ms-swift + Qwen3-VL family, and decisively: **start with the TAL head, not the VLM** (boundary precision is the product; the head is already de-risked on your corpus at $0 local). First two experiments = M0.4 harness + M0.2 grow to n>=5.
- **Q2 (conversational-QA):** Punt the feature, build the bookkeeping now. Architecture is extract-once -> event store -> structured text-to-SQL query layer; most of the store already exists in your SQLite schema. Reject end-to-end VLM-per-turn.
- **Q3 (multisport):** Doesn't change model/framework; scales data/compute/storage and reopens the per-sport clean-detector licensing front. Flagged the RacketVision refutation ? single-model multisport *temporal* generalization is now an open hypothesis, not proven.
- **Q4 (language):** Python everywhere; second language is a deferred, layer-local on-device runtime shell behind the export boundary, never a hot-path split.

I also surfaced the honest caveats (LOVO ~0.583 is far from "very high"; WASB C3 provenance; forbidden Gemini-CoT data; unverified video-GRPO VRAM) and the licensing watchlist, and noted that sequencing discipline holds ? this is strategy, not a go-build order.